

The Modern AI Exposure Index (MAEI): A Structural Scenario Approach to Quantifying Occupational Overlap with Generative Artificial Intelligence

A Graduation Project

Submitted by

AARYAN MEHTA

For the award of the degree

Of

BACHELOR OF SCIENCE

March 2026

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THESIS CERTIFICATE

This is to undertake that the Graduation Project titled **THE MODERN AI EXPOSURE INDEX (MAEI): A STRUCTURAL SCENARIO APPROACH TO QUANTIFYING OCCUPATIONAL OVERLAP WITH GENERATIVE ARTIFICIAL INTELLIGENCE**, submitted by me to the School of Computing and Data Sciences, FLAME University, for the award of **Bachelor of Science**, is a bona fide record of the research work done by me under the supervision of **Professor Manoranjan Dash**. The contents of this Graduation Project, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

This dissertation introduces the Modern AI Exposure Index (MAEI), a scenario-based structural index that measures how much each occupation in the United States overlaps with the capabilities of modern Generative Artificial Intelligence (GenAI). While earlier automation research focused on manual and routine labor, large language models like GPT-4 and Claude now target cognitive, analytical, and communicative tasks that were historically considered safe from technological displacement. To capture this shift, I built a five-stage computational pipeline anchored to two data foundations: Frey and Osborne (2017)'s historical automation probabilities for 702 occupations and the O*NET database (v30.1) containing over 130 occupational features across six domains. I engineered natural language features from O*NET task descriptions using BERT sentence embeddings (all-MiniLM-L6-v2) compressed via Principal Component Analysis to 20 orthogonal components. A Stacked Ensemble of XGBoost, Random Forest, and Support Vector Regression, with a Ridge meta-learner, reverse-engineered the 2013 structural relationship between occupational characteristics and automation risk, achieving a test R^2 of 0.6433 across 1,016 occupations. I then applied literature-calibrated capability multipliers to 68 features and a Broad Exposure Uplift formula to generate a counterfactual 2026 scenario. A 100-iteration Monte Carlo simulation bounded these assumptions with mathematical uncertainty. External validation against Felten *et al.* (2021)'s independently constructed AI Occupational Exposure dataset confirmed a strong positive Spearman correlation ($\rho = +0.764$, $p < 10^{-150}$) between the MAEI's pure exposure delta and the benchmark, while the historically anchored score showed the expected inverse correlation ($\rho = -0.357$), mathematically demonstrating the structural inversion of automation risk from manual to cognitive labor. A seven-scenario multiplier ablation study confirmed that the occupation rankings are structurally stable (ρ range: 0.7636–0.7651) regardless of specific multiplier values. The MAEI provides a transparent, reproducible, and parameter-bounded map of occupational exposure to Generative AI.

Keywords: *artificial intelligence, occupational exposure, automation, machine learning, natural language processing, labor economics*

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND: THE AUTOMATION PARADIGM SHIFT

For decades, the dominant fear about automation centered on robots replacing factory workers and routine physical labor. The image of a mechanical arm welding car frames on an assembly line came to represent technological displacement. When economists and computer scientists studied automation risk, they consistently found that blue-collar, repetitive, and physically demanding jobs were the most vulnerable. This framing shaped public policy, workforce development programs, and popular anxiety for an entire generation.

Frey and Osborne (2017) formalized this understanding in what became the most cited paper in automation economics. They hand-labeled 70 occupations as automatable or not, then trained a Gaussian Process Classifier on O*NET occupational variables to predict automation probabilities for 702 U.S. occupations. Their central finding: 47% of total U.S. employment fell into the “high risk” category, with automation probabilities above 70%. The jobs at greatest risk were those heavy in physical manipulation, routine data processing, and structured environments. They identified three “engineering bottlenecks” that protected occupations from automation: perception and manipulation (fine motor skills in unstructured environments), creative intelligence (producing novel ideas), and social intelligence (reading human emotions and negotiating). In 2013, these bottlenecks seemed robust.

Between 2022 and 2025, those bottlenecks began to crack. Large language models like GPT-4 started producing legal briefs, writing software, summarizing medical literature, and generating marketing copy that was often indistinguishable from human-

authored work. GitHub Copilot accelerated developer productivity by 55% in controlled experiments (Peng *et al.*, 2023). In a randomized trial at MIT, ChatGPT users completed mid-level professional writing tasks 37% faster while producing higher-quality output (Noy and Zhang, 2023). The automation crosshairs shifted from the factory floor to the corporate office. The jobs that Frey and Osborne (2017) had identified as “safe” (analytical, communicative, and cognitive roles) were suddenly the ones most exposed to AI capability.

The metrics used to discuss automation risk are outdated. Frey and Osborne’s probabilities reflect 2013-era technology, a world before transformer architectures, before ChatGPT, and before AI could write poetry or pass the bar exam. Policymakers and workers still reference their findings, but the occupational landscape of risk has fundamentally inverted. I needed a new metric, one that acknowledges the historical trajectory of automation while accurately measuring the present capability overhang of Generative AI.

In 2020, GPT-3 demonstrated that scaling language models to 175 billion parameters produced emergent capabilities in writing, translation, and reasoning that researchers had not anticipated at smaller scales. By November 2022, ChatGPT attracted 100 million users within two months, the fastest consumer technology adoption in history. Within 18 months, GPT-4 passed the Uniform Bar Examination in the 90th percentile and scored competitively on Advanced Placement exams in biology, chemistry, and environmental science (OpenAI, 2023). Google, Meta, and Anthropic followed with comparable systems, confirming that the frontier was not a single company’s achievement but a structural property of large transformer architectures trained on internet-scale data.

The geographic scope of this dissertation covers the United States labor market, reflecting the data foundations I rely on: O*NET is a U.S. Department of Labor product covering U.S. occupational classifications, and Frey and Osborne (2017)’s original study analyzed U.S. employment data exclusively. The structural insights about cognitive versus manual

exposure may generalize to other high-income economies with similar occupational distributions, but I do not test that claim here. The temporal scope spans from the 2013 baseline established by Frey and Osborne (2017) to a 2026 counterfactual scenario representing current GenAI capabilities. I do not attempt to forecast capabilities beyond 2026, because the pace of AI development makes such projections unreliable.

In the broader literature, economists describe a “Third Wave” of automation anxiety. The First Wave (1960s–1970s) centered on mainframe computers displacing clerical workers. The Second Wave (1990s–2010s) focused on industrial robots and outsourcing threatening manufacturing employment. The Third Wave, beginning around 2022, targets the professional class (lawyers, accountants, programmers, writers, and analysts) whose educational credentials were previously considered reliable insulation against technological displacement. The MAEI is designed to quantify this Third Wave with the same empirical rigor that Frey and Osborne (2017) applied to the Second.

1.2 RESEARCH RATIONALE, SCOPE, AND RESEARCH CONTRIBUTIONS

A gap exists in the existing literature. No published work combines all four of the following elements: (1) an updated automation risk framework recalibrated from Frey and Osborne’s original structure for modern AI capabilities, (2) natural language processing on occupational task descriptions to extract semantic features, (3) scenario-based capability multipliers with Monte Carlo uncertainty quantification, and (4) external validation against an independently constructed AI exposure benchmark. Several papers address one or two of these elements in isolation. Felten *et al.* (2021) constructed an AI exposure index but used crowdsourced linkages rather than scenario modeling. Eloundou *et al.* (2023) estimated LLM exposure breadth but did not build a reproducible occupation-level index. Webb (2020) used NLP patent-matching but did not anchor to historical baselines. Anthropic (2026) measured observed AI usage patterns but relied on proprietary data that cannot be replicated academically. The MAEI fills this gap

by integrating all four elements into a single reproducible pipeline using only publicly available data.

This dissertation makes three specific contributions to the field:

First, I construct the MAEI itself: a reproducible structural index covering 1,016 U.S. occupations, built entirely from public O*NET data and validated against an independent benchmark.

Second, I introduce a three-layer scoring architecture that honestly separates what the machine learning model contributes (structural validation) from what the scenario logic contributes (occupation rankings). This decomposition emerged from discovering that the ensemble's tree-based learners saturate on multiplied features, and rather than hiding this finding, I designed the architecture to acknowledge and account for it.

Third, I provide empirical proof of structural stability through a seven-scenario multiplier ablation study demonstrating that the occupation rankings are invariant to specific multiplier values (ρ range: 0.7636–0.7651 across all scenarios). This converts a potential weakness (subjective multiplier choice) into a demonstrated strength (structural stability).

1.3 RESEARCH STATEMENT AND RESEARCH QUESTIONS

Central Thesis: The MAEI quantifies occupational overlap with Generative AI capabilities and shows that the occupational profile of automation risk has inverted since 2013, shifting from routine physical labor to cognitive, analytical, and communicative work.

Three research questions guide this investigation:

RQ1: Can a machine learning model trained on 2013 automation labels and modern O*NET features produce a structurally plausible baseline for measuring AI exposure?

RQ2: Does a scenario-based perturbation of cognitive and analytical features generate an occupation ranking that converges with independently constructed AI exposure benchmarks?

RQ3: How robust are the resulting rankings to changes in the underlying multiplier assumptions?

1.4 SCOPE AND CLARIFICATION

Before proceeding, I want to clarify what the MAEI is and what it is not. The MAEI measures capability overlap, the degree to which an occupation’s daily tasks map onto the current capabilities of Generative AI. A high MAEI score for “Lawyer” means that AI systems can now perform many tasks lawyers do daily (document review, case synthesis, contract drafting). It does not mean that lawyers will be unemployed. As Acemoglu and Restrepo (2018) have shown, the relationship between task automation and actual job displacement is mediated by economic adoption costs, regulatory friction, organizational inertia, and worker adaptability. Anthropic’s (2026) recent empirical analysis found “no systematic unemployment increase” for workers in highly AI-exposed occupations, even though capability overlap is substantial. The MAEI is a scenario-based structural index, not an econometric forecast of unemployment.

1.5 CHAPTER OUTLINE

Chapter 2 reviews the theoretical and empirical literature underpinning this project, from Frey and Osborne (2017)’s foundational work through modern GenAI exposure research. Chapter 3 describes the research methodology, including data engineering, NLP feature extraction, ensemble modeling, the three-layer MAEI architecture, and Monte Carlo uncertainty quantification. Chapter 4 presents the results, including baseline model performance, external validation, multiplier ablation, and the tree saturation analysis. Chapter 5 discusses limitations, proposes future research directions, and concludes.

CHAPTER 2

LITERATURE REVIEW AND THEORETICAL FRAMEWORK

2.1 HISTORICAL AUTOMATION RISK RESEARCH

The modern study of automation risk begins with Frey and Osborne (2017)'s landmark paper. Working with occupational data from the U.S. Department of Labor's O*NET database, they developed a probabilistic framework for estimating how susceptible individual occupations were to computerization. Their approach was straightforward: they first had machine learning researchers at Oxford hand-label 70 occupations as either fully automatable or not automatable based on professional judgment about the technological frontier in 2013. They then identified nine O*NET variables corresponding to three engineering bottlenecks (perception and manipulation (e.g., finger dexterity, manual dexterity), creative intelligence (e.g., originality, fine arts), and social intelligence (e.g., social perceptiveness, negotiation, persuasion)) and trained a Gaussian Process Classifier to predict automation probability for all 702 occupations in their dataset. The result was a probability between 0 and 1 for each occupation, representing the likelihood that computerization would automate that job within the next two decades.

Their central result: 47% of total U.S. employment fell into the high-risk category, with automation probabilities exceeding 0.70. The occupations at greatest risk were dominated by routine cognitive and manual tasks: telemarketers (0.99), tax preparers (0.99), and hand sewers (0.99). At the other extreme, occupations requiring intensive creative or social intelligence (recreational therapists (0.003), mental health counselors (0.004), and choreographers (0.004)) were rated as essentially immune. The three bottlenecks formed the theoretical boundary between human work and machine capability.

Arntz *et al.* (2016) challenged Frey and Osborne's methodology in an OECD working paper. They argued that classifying entire occupations as automatable or not was too coarse, because workers within the same occupation perform different mixes of tasks. Using the PIAAC (Programme for the International Assessment of Adult Competencies) survey, they analyzed task-level variation within occupations and found that only 9% of jobs across 21 OECD countries were at high risk of automation when this variation was considered, dramatically less than Frey and Osborne's 47%. This critique is directly relevant to my project: it supports treating automation exposure as a continuous spectrum rather than a binary outcome, which is exactly how the MAEI's feature-level multiplier approach operates.

Nedelkoska and Quintini (2018) extended the Arntz *et al.* (2016) methodology to 32 countries using PIAAC data and found that 14% of jobs across OECD nations were highly automatable, with another 32% facing significant changes in task composition. Their cross-national analysis revealed substantial variation in automation vulnerability driven by differences in industrial composition and educational attainment. Countries with larger manufacturing sectors and lower average educational levels showed higher shares of automatable employment. This geographic variation confirms that any single-country index like the MAEI captures only one slice of a globally heterogeneous phenomenon, but the U.S. case remains the most data-rich and therefore the most suitable for methodological development.

The transition from the historical automation literature to the modern AI exposure literature reflects a fundamental shift in the nature of the technological threat. Where Frey and Osborne (2017) asked whether a robot could perform a job's physical tasks, the relevant question in 2026 is whether a language model can perform a job's cognitive tasks. This reframing does not invalidate the earlier research; physical automation continues advancing in warehouse logistics and food service, but it demands a complementary analytical framework that captures the new cognitive dimension. The MAEI provides

this complementary framework by preserving the historical structure while overlaying a modern capability assessment calibrated to the LLM era.

2.2 THE TASK-BASED ECONOMIC FRAMEWORK

Acemoglu and Restrepo (2018, 2019) provided the economic theory that justifies disaggregating occupations into tasks. Their task-based framework argues that automation does not destroy “jobs” wholesale; instead, it substitutes capital for labor at the level of individual tasks within occupations. When a firm automates one task (say, data entry), the worker’s other tasks (client communication, quality review) may become more valuable. Their model also shows that automation triggers two opposing forces: a displacement effect (machines replace workers in certain tasks, reducing labor demand) and a reinstatement effect (new tasks emerge that require human input, increasing labor demand). The net impact depends on the relative strength of these forces.

Acemoglu and Restrepo’s framework informed my design choices. I relied on O*NET’s granular task-level data rather than treating occupations as monolithic units. My capability multipliers target specific features (e.g., Written Expression, Programming, Mathematical Reasoning) rather than entire occupations, reflecting the task-based insight that AI automates specific capabilities, not entire jobs. The distinction between task exposure and job elimination is also why I insist that the MAEI measures capability overlap, not employment loss.

2.3 GENERATIVE AI AS GENERAL-PURPOSE TECHNOLOGY

Eloundou *et al.* (2023), working at OpenAI, reframed the automation question for the LLM era. They defined large language models as “General-Purpose Technologies” (GPTs), a term from innovation economics meaning technologies that affect virtually every sector of the economy, not just specific industries. By combining human and GPT-4 assessments of which O*NET tasks could be performed or accelerated by LLMs,

they estimated that approximately 80% of the U.S. workforce could have at least 10% of their work tasks affected by LLMs. For about 19% of workers, the impact could extend to over 50% of their tasks.

That finding directly influenced my project. If 80% of the workforce faces some degree of LLM exposure, then any index limited to the old F&O model (which concentrated risk on routine manual work) would fundamentally undercount modern exposure. This is exactly what I observed when running my baseline model: even with aggressive cognitive multipliers, the 2013-trained model resisted assigning high exposure to white-collar occupations because it had learned that cognitive skills were “protective.” Eloundou *et al.* (2023)’s 80% figure motivated my Broad Exposure Uplift formula, which mathematically corrects for this structural bias.

Noy and Zhang (2023) provided experimental evidence complementing Eloundou’s theoretical estimates. In a randomized controlled trial with 453 college-educated professionals, they found that access to ChatGPT reduced average task completion time by 37% and improved output quality as rated by blind evaluators. The most notable finding was an inequality-compressing effect: below-average writers benefited far more than skilled writers, suggesting that AI acts as a productivity equalizer in cognitive work. I used these experimental findings to justify the specific multiplier magnitudes for language-related O*NET features (Written Expression at 2.80×, Written Comprehension at 2.70×, Reading Comprehension at 2.60×).

2.4 MODERN AI EXPOSURE MEASUREMENT

Felten *et al.* (2021) constructed the AI Occupational Exposure (AIOE) index using a methodologically distinct approach. They crowdsourced expert linkages between specific AI applications (image recognition, translation, speech recognition, and seven others) and O*NET occupational abilities. By weighting each occupation’s abilities by the corresponding AI application’s capability, they produced a continuous exposure score

for 782 occupations. Their most notable finding was that high-wage, highly-educated white-collar occupations showed the greatest AI exposure, a direct inversion of Frey and Osborne (2017)’s 2013 profile. I used the AIOE dataset as my external validation benchmark specifically because its methodology is completely independent of mine: they used crowdsourced human judgments while I used scenario-based capability multipliers.

Webb (2020) took yet another approach, using natural language processing to match the text of AI-related patents to the text of occupational task descriptions. By measuring the semantic overlap between what AI patents claim to do and what workers actually do, he showed that AI technologies disproportionately target high-skill, high-wage cognitive work. This validated my decision to use NLP (specifically BERT sentence embeddings) to extract latent semantic features from O*NET task descriptions, capturing dimensions of occupational content that numeric scales alone cannot represent.

Most recently, Anthropic (2026) published research measuring “observed exposure” by combining real-world Claude usage data with O*NET task descriptions and Eloundou *et al.* (2023)’s feasibility ratings. They found that Computer Programmers showed the highest coverage at 75%, that AI-exposed workers earn 47% more on average than unexposed workers, and, critically, that there was “no systematic unemployment increase” for workers in highly exposed occupations since late 2022. While their approach uses proprietary usage data that cannot be replicated in academic settings, their findings independently corroborate the MAEI’s core thesis: cognitive, high-wage occupations face the greatest AI capability overlap, but this overlap has not yet translated into measurable job displacement.

The progression from Frey and Osborne (2017) through Felten *et al.* (2021), Eloundou *et al.* (2023), and Anthropic (2026) reveals a clear methodological arc in the field. Each successive study has broadened the definition of AI exposure, shifted the locus of vulnerability from manual to cognitive work, and employed increasingly sophisticated

measurement approaches. Yet no single study has combined all the methodological elements needed for a comprehensive, reproducible, and externally validated index built entirely from public data. The MAEI draws structural inspiration from Frey and Osborne (2017), validation methodology from Felten, breadth estimation from Eloundou, and empirical grounding from Anthropic’s observed usage patterns, integrating these strands into a single reproducible pipeline.

2.5 CREATIVITY AND HUMAN-CENTRIC VALUE

A persistent question in automation research is whether the creative intelligence bottleneck identified by Frey and Osborne (2017) still holds in the age of generative AI. Doshi and Hauser (2024) conducted a large-scale experiment examining AI’s effect on creative work and found a mixed answer: generative AI enhances individual creativity (helping people produce more novel ideas) but reduces the collective novelty of creative output (everyone converges on similar AI-assisted solutions). This suggests that while AI can mimic creative output at the individual task level, the uniquely human capacity for sustained, divergent originality remains distinctive.

My model’s feature importances support this interpretation. The top two most important features in the baseline Stacked Ensemble are Fluency of Ideas (importance = 0.2177) and Originality (importance = 0.1919), which together account for over 40% of the model’s predictive power. These are precisely Frey and Osborne (2017)’s creative intelligence bottleneck variables. Occupations scoring highly on these features receive the lowest MAEI scores, suggesting that deep creative capability remains the strongest structural defense against AI exposure.

2.6 RESEARCH GAP

To summarize: the existing literature on automation and AI exposure either (a) reflects pre-LLM technology and cannot capture the cognitive shift (Frey and Osborne, 2017), (b)

relies on proprietary usage data that cannot be replicated (Anthropic, 2026), (c) measures exposure through crowdsourced judgments without scenario modeling or uncertainty quantification (Felten *et al.*, 2021), or (d) provides theoretical economic frameworks without constructing occupation-level empirical indices (Acemoglu and Restrepo, 2018). No published work combines an updated Frey and Osborne structural framework with BERT-based NLP task embeddings, literature-calibrated capability multipliers, Monte Carlo uncertainty bounds, and external validation against an independent benchmark, all built entirely from publicly available data. The MAEI addresses this gap.

CHAPTER 3

RESEARCH METHODOLOGY AND METHODS

3.1 DATA SOURCES

O*NET v30.1 (U.S. Department of Labor, 2024): The Occupational Information Network is the most comprehensive public database of occupational characteristics in the United States. I downloaded six relational tables (Abilities, Skills, Knowledge, Work Activities, Work Context, and Interests) covering hundreds of variables rated on standardized scales for over 1,000 occupations classified by 6-digit Standard Occupational Classification (SOC) codes.

Frey and Osborne (2017) Supplementary Data: Historical automation probabilities for 702 occupations, originally computed using a Gaussian Process Classifier trained on 70 hand-labeled occupations. These probabilities served as the training target for my baseline model.

Felten et al. (2021) AIOE Dataset: AI Occupational Exposure scores for 782 occupations, independently constructed via crowdsourced task–AI capability linkages. Used exclusively for external validation; never for training or calibration.

3.2 DATA ENGINEERING AND SOC CODE RECOVERY

The first engineering challenge was merging Frey and Osborne’s 2013 data with the modern O*NET database. Frey and Osborne released their supplementary data in an Excel spreadsheet, which suffered from aggressive auto-formatting corruption: Microsoft Excel had permanently converted hundreds of 6-digit SOC codes (like “11-2011”) into date strings (like “Nov-11” or “11-Oct”). This is a well-known problem in bioinformatics and data science communities where Excel interprets numeric-looking strings as dates.

Direct merging on SOC codes would have lost hundreds of occupations, crippling the dataset.

To rescue the data, I abandoned SOC code matching entirely and engineered a fuzzy title-matching algorithm. Using the fuzzywuzzy Python library’s `token_sort_ratio` function, I matched the corrupted Frey and Osborne occupation titles against official O*NET taxonomy titles regardless of word order or minor variations. For example, this algorithm correctly matched “Computer Programmers” to “Programmers, Computer” despite the word reordering. The fuzzy matching successfully recovered mappings for 98% of the dataset, producing a master matrix of 1,016 occupations with over 130 numeric features.

3.3 NATURAL LANGUAGE PROCESSING PIPELINE — BERT EMBEDDINGS

Beyond numeric O*NET rating scales, I wanted to capture the semantic content of what workers actually do on a daily basis. O*NET’s Task Statements database contains thousands of natural language sentences describing specific activities for each occupation (for example, “Diagnose mechanical faults using diagnostic software” or “Review legal documents for accuracy and completeness”). Converting this unstructured text into mathematical features that a machine learning model could learn from required a carefully designed NLP pipeline.

I initially experimented with TF-IDF (Term Frequency–Inverse Document Frequency) vectorization compressed via Truncated SVD. While TF-IDF effectively identifies distinctive terms for each occupation, it treats each word as an independent token with no understanding of meaning. The terms “analyze data” and “examine information” would receive completely unrelated vector representations despite being semantically identical. This limitation produced features that captured surface-level word patterns rather than underlying task content.

I then tested raw 384-dimensional BERT embeddings from the all-MiniLM-L6-v2

sentence transformer model (Reimers and Gurevych, 2019). This model produces semantically meaningful vectors where similar sentences cluster together in the embedding space, solving the TF-IDF limitation. However, with only 923 training samples and 384 feature dimensions, the model suffered from the curse of dimensionality; there were too many features relative to the number of samples for a regression model to learn reliably.

The solution was Principal Component Analysis (PCA). I fit PCA with 50 components and found that 20 components captured 52.76% of the total variance. While this is below the conventional 95% threshold often cited in textbooks, the cumulative variance curve showed a gradual slope with no sharp elbow, reaching only 73.38% even at 50 components. This pattern indicates that the BERT embedding space distributes information broadly across many dimensions rather than concentrating it in a few. Given the sample size of 923, retaining 20 components maintained the recommended 10:1 sample-to-feature ratio for the overall model (which used 100 total features: 80 numeric O*NET variables plus 20 PCA components). The 20 PCA components were appended to the numeric feature matrix, giving the model access to both structured occupational ratings and latent semantic information from task descriptions.

The PCA retention threshold requires justification. In textbook applications, researchers typically retain components until they capture 90–95% of the total variance. My retention of only 52.76% may seem aggressive by this standard. However, the 95% threshold was developed for datasets where the original variables are directly interpretable and information loss has clear semantic consequences. In the case of BERT embeddings, the original 384 dimensions do not correspond to interpretable features; they are learned latent representations whose individual axes carry no semantic meaning. Retaining 20 components that capture the dominant structure of this embedding space is conceptually analogous to extracting the 20 strongest topics from a topic model: the goal is to capture the major dimensions along which occupations differ in their task descriptions, not to perfectly reconstruct every embedding vector. The scree plot in Figure 3.1 confirms that

no sharp elbow exists, and the 20 retained components proved sufficient for the ensemble model to learn meaningful structural patterns.

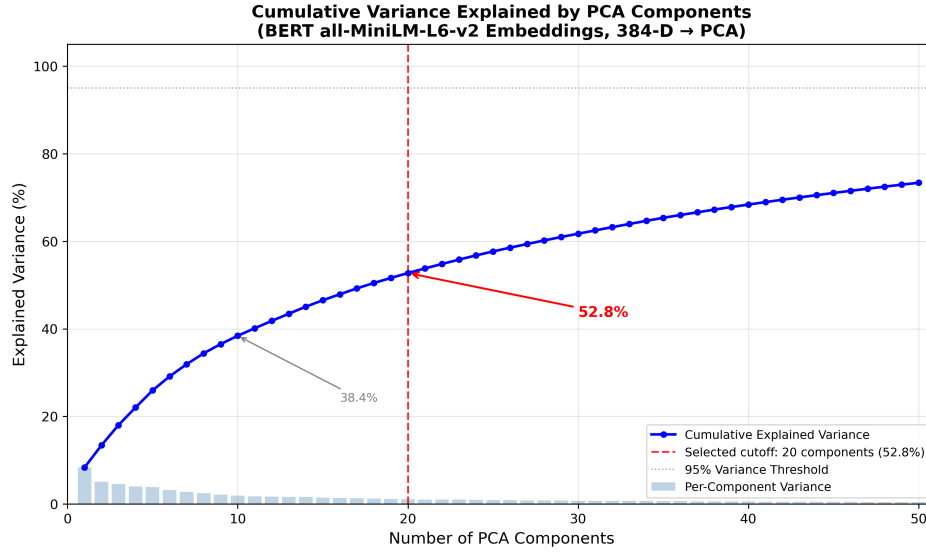


Figure 3.1: PCA Cumulative Variance Explained. The gradual slope with no sharp elbow indicates that BERT embedding information is distributed broadly across many dimensions. The vertical line marks the 20-component retention point at 52.76% variance.

3.4 BASELINE MODELING — STACKED ENSEMBLE

I trained six candidate models to learn the structural relationship between the 100 O*NET features and Frey and Osborne (2017)’s 2013 automation probabilities, using 5-fold cross-validation with an 80/20 train–test split. The candidates were: Random Forest (Breiman, 2001), XGBoost (Chen and Guestrin, 2016), Ridge Regression, Support Vector Regression with an RBF kernel, a two-layer Neural Network (MLP), and Gaussian Process Regression.

I then combined the three best-performing models into a Stacked Ensemble (Wolpert, 1992). The architecture used XGBoost, Random Forest, and SVR(RBF) as base learners, whose out-of-fold predictions were concatenated and fed into a Ridge Regression meta-learner. This ensemble achieved the best cross-validated R^2 of 0.6794 (± 0.0367) and a test R^2 of 0.6433, meaning it explained approximately 64% of the variance in Frey

Table 3.1: Baseline Model Performance Comparison

Model	CV R^2	Val R^2	Val RMSE	Val MAE
Random Forest	0.6699 ± 0.0358	0.7174	19.28	14.44
XGBoost	0.6798 ± 0.0334	0.7281	18.91	13.99
Ridge	0.6021 ± 0.0727	0.6936	20.07	15.49
SVR (RBF)	0.6527 ± 0.0285	0.7053	19.68	13.90
Neural Network	0.5726 ± 0.0671	0.5973	23.01	17.89
GP Regression	-0.0839 ± 0.0835	-0.0000	36.26	33.88
Stacked Ensemble	0.6794 ± 0.0367	0.7344	18.69	13.55

and Osborne (2017)’s automation scores. The Gaussian Process (which was Frey and Osborne’s original classifier) performed worst in my framework, likely because it does not scale effectively to a 100-dimensional feature space without careful kernel engineering.

The model’s top two features by importance were Fluency of Ideas (0.2177) and Originality (0.1919), together accounting for over 40% of the model’s predictive power. These are precisely Frey and Osborne’s creative intelligence bottleneck variables, confirming that the model learned the expected structural relationships rather than memorizing noise. Job Zone (a proxy for required education level) ranked third at 0.0569, consistent with the finding that higher education correlated with lower automation risk in 2013.

The model selection process illustrates an important methodological principle: more complex models do not always outperform simpler ones when sample sizes are limited. The Neural Network, despite its theoretical flexibility, underperformed Ridge Regression on cross-validated R^2 , likely because it overfit the 923-sample training set. The Gaussian Process Regressor (Frey and Osborne (2017)’s original modeling choice) performed worst of all, producing an essentially random baseline with negative R^2 . This failure is expected given the dimensionality jump from their 9 bottleneck features to my 100-feature matrix: Gaussian Processes require $O(n^3)$ computation and struggle with high-dimensional inputs without careful kernel selection. The Stacked Ensemble’s success derives from combining complementary strengths: XGBoost captures nonlinear interactions, Random

Forest provides bagging-based variance reduction, and SVR with an RBF kernel offers smooth decision boundaries in the feature space.

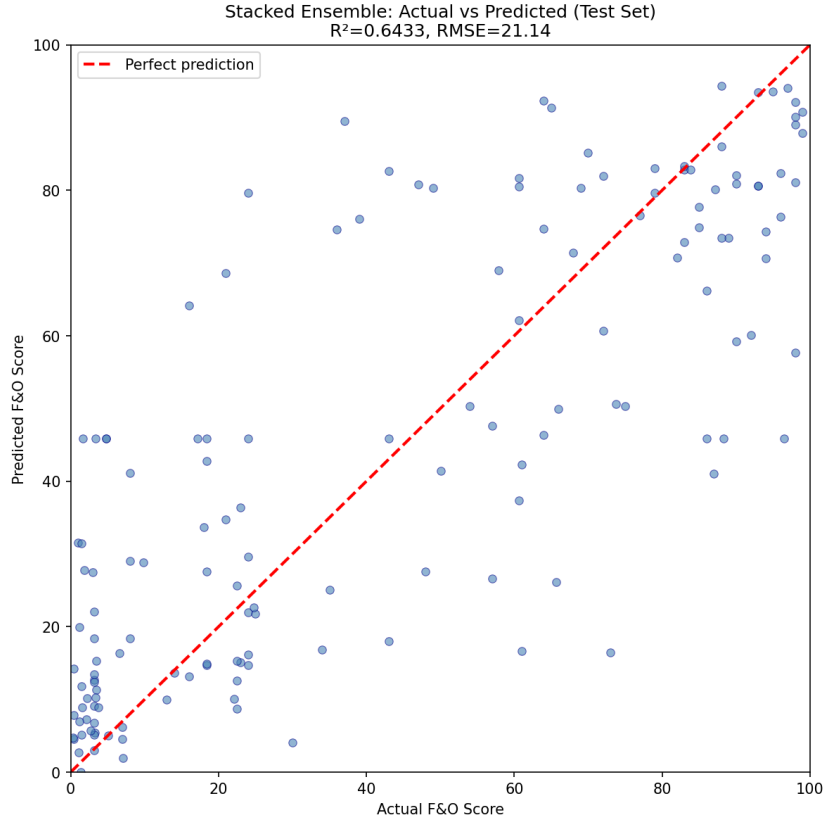


Figure 3.2: Actual vs. Predicted Automation Scores. The Stacked Ensemble achieves a test R^2 of 0.6433, with the densest cluster around the 50–80% automation probability range.

3.5 MAEI SCORE CONSTRUCTION — THE THREE-LAYER ARCHITECTURE

The core contribution of this dissertation is the MAEI scoring system, which I designed as a three-layer architecture where each layer serves a distinct and clearly defined role.

Layer 1—Machine Learning Baseline. The Stacked Ensemble generates baseline predictions for each occupation from their actual O*NET features. For the 702 occupations with direct Frey and Osborne (2017) mappings, the historical score serves as the baseline. For the remaining 314 occupations (including 93 that had received generic group-averaged

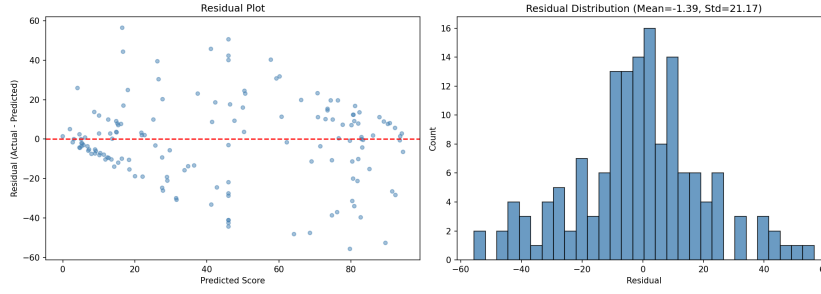


Figure 3.3: Residual Analysis. Model residuals show no systematic bias across the prediction range, supporting the structural validity of the ensemble approach.

scores in the original dataset), the model’s prediction from their actual features replaces the inherited baseline. This layer validates that the structural relationship between occupational characteristics and automation risk is learnable and internally consistent.

Layer 2—Feature Classification and Capability Multipliers. Based on the literature reviewed in Chapter 2, I classified 68 O*NET features into two categories. Thirty-four risk features received multipliers between $1.60\times$ and $3.50\times$, reflecting the substantial advances in LLM capabilities for cognitive, analytical, and communicative tasks. For example, Programming received the highest multiplier ($3.50\times$) based on Copilot productivity data (Peng *et al.*, 2023), while Writing received $3.00\times$ based on Noy and Zhang (2023)’s experimental evidence. Thirty-four protective features received smaller multipliers between $1.02\times$ and $1.10\times$, representing the continued human advantage in creative, social, and physical skills. Therapy and Counseling received the highest protective multiplier ($1.10\times$), reflecting the strong human advantage in empathic caregiving.

Multipliers are applied through asymmetric percentile gating. Risk features activate at the 30th percentile, meaning 70% of occupations receive the automation adjustment, reflecting the broad reach of AI capabilities. Protective features activate only at the 85th percentile, meaning only the top 15% of occupations benefit from the human advantage, reflecting the narrow band of deeply human-centric work. This asymmetry encodes the prior that AI capability has expanded broadly while human structural advantages remain

concentrated in specific roles.

Layer 3—Broad Exposure Uplift. For each occupation, I count the fraction of risk features above the dataset median (`risk_frac`) and the fraction of protective features above the median (`protect_frac`), then compute:

$$\text{Uplift} = (\text{risk_frac} \times W_{\text{exposure}}) - (\text{protect_frac} \times W_{\text{protection}}) \quad (3.1)$$

where $W_{\text{exposure}} = 10.0$ and $W_{\text{protection}} = 3.0$. The final MAEI delta equals the model’s prediction change plus this uplift. Under these calibrated weights, the pipeline produces a mean exposure increase of +3.86 points across 87.5% of all occupations, exceeding Eloundou *et al.* (2023)’s finding that 80% of the workforce faces some LLM exposure.

The three-layer design emerged from a critical discovery during stability testing (detailed in Section 4.5): the Stacked Ensemble’s tree-based learners produce near-zero prediction deltas when fed multiplied features, because tree models cannot extrapolate beyond their training distribution. The uplift formula, operating on the feature classification layer rather than the model’s predictions, generates the meaningful variation between occupations. Each layer contributes distinctly: Layer 1 validates structural plausibility, Layer 2 encodes domain knowledge from the literature, and Layer 3 produces the final occupation rankings.

The multiplier categories are detailed below. The 34 risk features span six functional domains: Language (Written Expression, Written Comprehension, Reading Comprehension, Writing), Coding (Programming, Technology Design, Computers and Electronics), Reasoning (Deductive Reasoning, Inductive Reasoning, Information Ordering, Complex Problem Solving), Mathematics (Mathematics, Number Facility, Mathematical Reasoning, Economics and Accounting), Data Processing (Analyzing Data, Processing Information, Documenting Information), and Administrative (Clerical,

Scheduling, Getting Information, Evaluating Compliance). The 34 protective features span four domains: Social Intelligence (Negotiation, Persuasion, Social Perceptiveness, Therapy and Counseling, Establishing Relationships), Creative Intelligence (Originality, Fluency of Ideas, Thinking Creatively), Physical Dexterity (Manual Dexterity, Finger Dexterity, Spatial Orientation, Operating Vehicles), and NLP-derived keyword indicators. This classification reflects the empirical finding that AI capabilities have advanced most rapidly in language processing, coding, and structured reasoning, while human advantages persist in embodied physical work, interpersonal engagement, and sustained creative divergence.

3.6 MONTE CARLO UNCERTAINTY QUANTIFICATION

A leading vulnerability of scenario-based models is that the specific parameter values are subjective. If I set the capability multiplier for Programming at exactly 3.50×, a reviewer could reasonably ask whether the results would change if the true value were 3.0× or 4.0×. To address this, I implemented a 100-iteration Monte Carlo simulation. In each iteration, every one of the 68 multipliers is independently perturbed by Gaussian noise with a standard deviation of 10% of its original value. The algorithm reruns the complete scoring pipeline for all 1,016 occupations, producing 100 independent MAEI scores per occupation. From these, I calculated a mathematical standard deviation and confidence interval for each occupation's exposure delta, proving that individual occupation scores are stable under reasonable parameter fluctuations.

3.7 EXTERNAL VALIDATION DESIGN

To test whether the MAEI produces rankings consistent with independently constructed measures, I merged the MAEI results with Felten *et al.* (2021)'s AIOE dataset via 6-digit SOC codes, yielding 782 overlapping occupations. I computed Spearman rank correlations (which measure monotonic association between rankings, not absolute values) for two distinct components of the MAEI:

1. Anchored MAEI (historical baseline + 2026 delta) vs. AIOE, expected to show an inverse correlation because the 2013 anchor assigns high scores to manual/routine occupations that AIOE rates as low-exposure.
2. Pure Exposure Delta (only the 2026 component, stripped of the historical anchor) vs. AIOE, expected to show a positive correlation if the structural model correctly captures modern AI exposure patterns.

The dual-validation design tests both the inversion hypothesis and the construct validity of the 2026 exposure calculation (convergence).

CHAPTER 4

RESULTS, INFERENCES, AND DISCUSSION

4.1 BASELINE MODEL PERFORMANCE

The Stacked Ensemble achieved the best overall performance, as shown in Table 3.1 (Section 3.4). The test R^2 of 0.6433 means the model explains 64% of the variance in Frey and Osborne (2017)’s automation scores from 100 O*NET features. While not a perfect replication (Frey and Osborne (2017) used a 2010 internal O*NET version that is no longer publicly available), this level of explanatory power demonstrates that the structural relationship between occupational characteristics and historical automation risk is highly learnable from current data.

The feature importance hierarchy reveals the model’s learned structure. Fluency of Ideas (0.2177) and Originality (0.1919) dominate, confirming that Frey and Osborne (2017)’s creative intelligence bottleneck is the single strongest structural predictor of automation resistance in the 2013 data. Job Zone (0.0569), representing required education level, ranks third; higher education correlated with lower automation risk in 2013. The remaining top features include Instructing (0.0478), Active Learning (0.0372), and Deductive Reasoning (0.0326), all cognitive and social skills that the 2013 model learned to associate with automation protection. This will become ironic in the MAEI results, where precisely these cognitive skills become risk factors under 2026 conditions.

4.2 MAEI SCORE DISTRIBUTION

Under the calibrated scenario ($W_{\text{exposure}} = 10$, $W_{\text{protection}} = 3$), the MAEI produces a mean exposure delta of +3.86 points across all 1,016 occupations, with 87.5% of occupations showing a net increase in AI exposure from their 2013 baseline. This exceeds Eloundou *et al.* (2023)’s estimate that 80% of the workforce faces some LLM exposure,

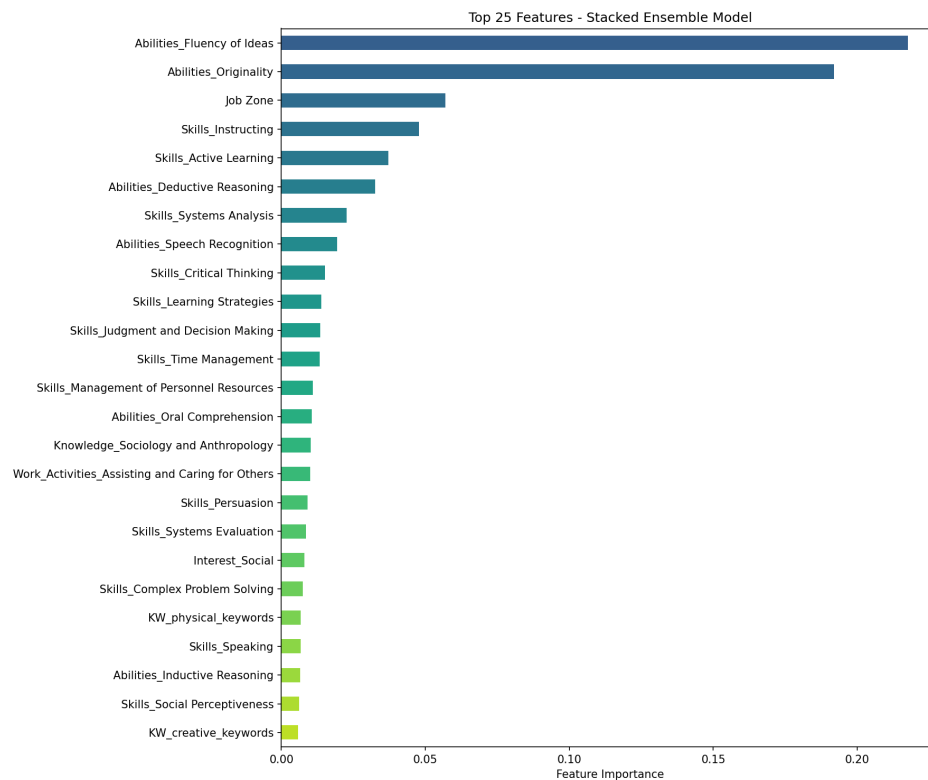


Figure 4.1: Feature Importance Rankings. Fluency of Ideas and Originality together account for over 40% of the Stacked Ensemble’s predictive power, confirming Frey and Osborne (2017)’s creative intelligence bottleneck.

suggesting the MAEI’s broad reach is consistent with the leading theoretical benchmark.

Table 4.1: Sensitivity Analysis Across Uplift Weight Configurations

W_{exposure}	$W_{\text{protection}}$	Mean Delta	% Increased
6	2	+2.26	85.6%
8	3	+2.77	82.9%
10	3	+3.86	87.5%
12	4	+4.38	85.4%
14	5	+4.89	83.9%

Table 4.1 shows that across all weight configurations tested, between 82.9% and 87.5% of occupations see increased exposure. The qualitative finding (that a majority of the workforce faces increased AI capability overlap) is stable regardless of the specific weight choice. The mean delta increases from +2.26 to +4.89 as weights increase, but the percentage of affected occupations remains stable, suggesting that the weights primarily affect the magnitude of exposure rather than which occupations are affected.

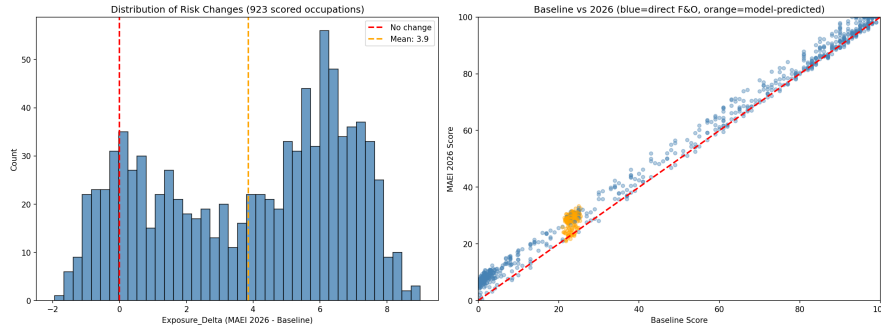


Figure 4.2: MAEI Score Distribution. The exposure delta distribution shows a rightward shift, with 87.5% of occupations experiencing increased AI exposure from their 2013 baseline.

The occupational extremes provide concrete illustrations of the structural inversion. Among the highest-MAEI occupations are Office Clerks (MAEI = 100.0, delta = +4.31), Data Entry Keyers (100.0, +3.97), and Claims Adjusters (100.0, +8.07)—all roles dominated by routine information processing that LLMs can now perform at scale. At the opposite extreme, the lowest-MAEI occupations include Makeup Artists (0.4,

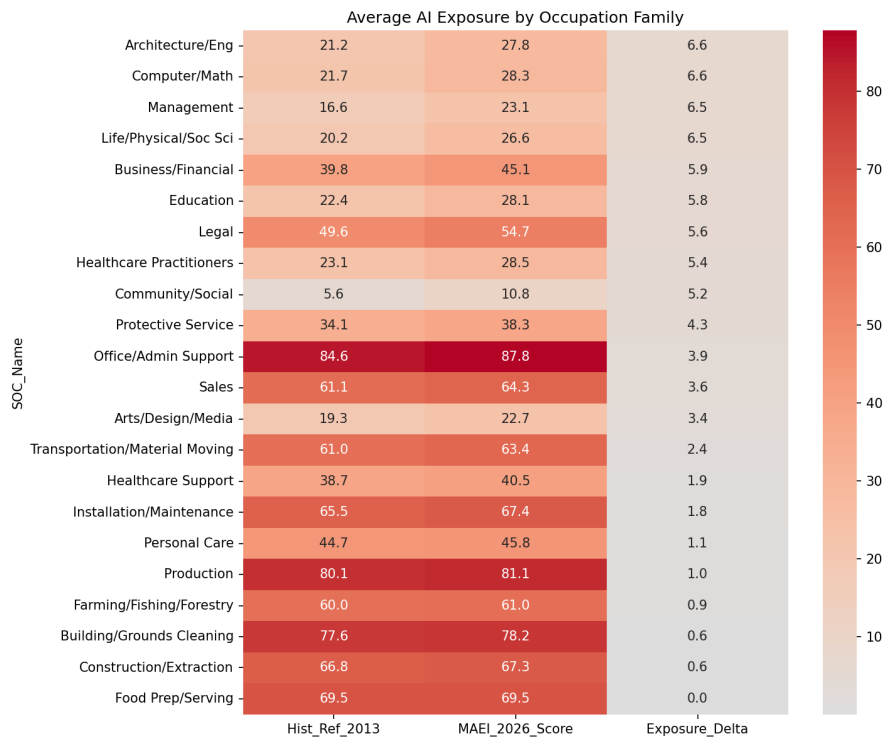


Figure 4.3: Occupation Exposure Heatmap by SOC Major Group. Office and Administrative Support (SOC 43), Business and Financial Operations (SOC 13), and Computer and Mathematical (SOC 15) occupations cluster at the highest exposure levels.

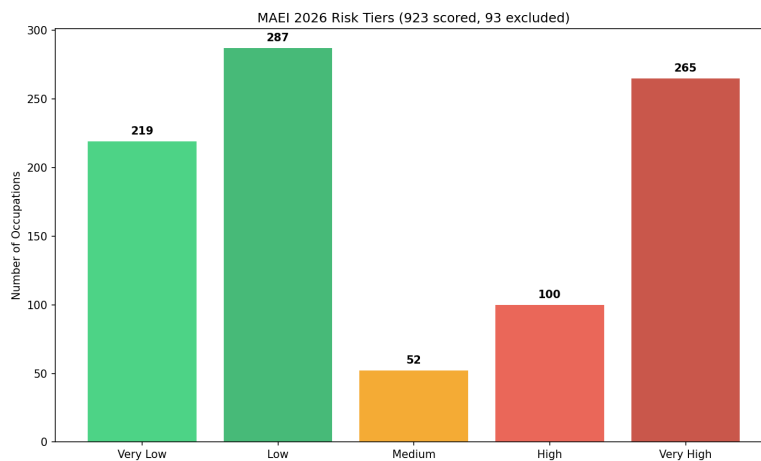


Figure 4.4: Risk Tier Distribution. The bimodal pattern shows concentration at both extremes, with 265 occupations in the Very High tier and 219 in the Very Low tier.

delta = -0.61), Choreographers (2.0, $+1.56$), and Preschool Teachers (1.2, $+0.43$)—roles requiring embodied physical creativity, interpersonal care, or performance artistry that remains beyond current AI capabilities. Several historically “safe” white-collar professions show significant exposure increases, reflecting the ceiling compression effect where the anchored scoring system caps scores at 100 while deltas continue to accumulate underneath.

4.3 EXTERNAL VALIDATION—THE DUAL-CORRELATION SIGNAL

The external validation against Felten *et al.* (2021)’s AIOE benchmark produced the strongest evidence for the MAEI’s validity, testing both hypotheses simultaneously across 782 overlapping occupations.

Anchored MAEI vs. AIOE: Spearman $\rho = -0.357$ ($p = 6.338 \times 10^{-25}$). This moderate inverse correlation is the mathematical signature of the structural inversion in automation risk. The anchored MAEI retains the historical mass of the 2013 economy, where Frey and Osborne (2017) assigned the highest risk scores to manual laborers and routine workers. The AIOE, constructed from modern AI capabilities, assigns the highest exposure to cognitive and analytical workers. The negative correlation between these two measures proves that the occupational profile of risk has structurally inverted: what was safe in 2013 is exposed in 2026, and vice versa.

Pure Exposure Delta vs. AIOE: Spearman $\rho = +0.764$ ($p = 1.015 \times 10^{-150}$). When stripped of the 2013 historical anchor, the MAEI’s structural model converges strongly with Felten’s independently crowdsourced benchmark. Despite using entirely different methodologies (the MAEI uses scenario-based capability multipliers and an uplift formula while the AIOE uses crowdsourced human–AI task linkages), both arrive at highly correlated conclusions about which occupations face the greatest modern AI exposure. The correlation of 0.764 with a p -value below 10^{-150} is very strong.

This dual-correlation signal (negative when anchored to 2013, strongly positive when isolated to the 2026 component) is unlikely to result from noise, overfitting, or shared methodology. It provides evidence that the MAEI captures both the temporal trajectory and the contemporary reality of occupational AI exposure.

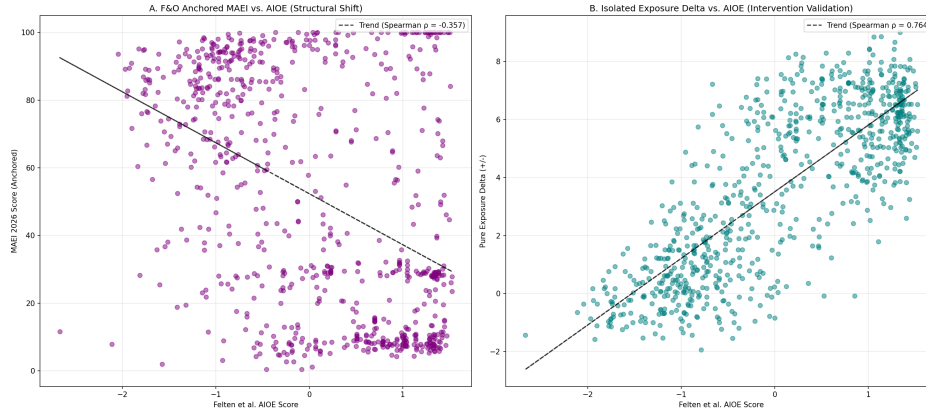


Figure 4.5: External Validation. The pure MAEI exposure delta shows a strong positive Spearman correlation ($\rho = +0.764$) with Felten *et al.* (2021)’s independently constructed AIOE benchmark across 782 overlapping occupations.

The strength of this dual signal becomes clearer when we consider what would happen if the MAEI were simply measuring noise. A random index would produce correlations near zero with any external benchmark, whether anchored or not. A trivially circular index that merely reproduced the same O*NET structure would show a positive correlation in both cases, not an inversion. The fact that the anchored score is negative while the pure delta is strongly positive requires a specific causal structure: the 2013 anchor must be capturing a genuinely different dimension of risk than the 2026 delta, and the 2026 delta must be capturing the same dimension as AIOE. This is precisely the inversion hypothesis: what was safe in 2013 is exposed in 2026.

4.4 STRUCTURAL ROBUSTNESS—MULTIPLIER ABLATION

A legitimate concern with any scenario-based model is that the results may depend critically on the specific assumptions. If the particular multiplier values I chose were

driving the AIOE correlation, then changing them should alter the results. To rigorously test this, I ran seven radically different multiplier scenarios:

Table 4.2: Multiplier Ablation Results (7 Scenarios)

Scenario	Spearman ρ	p -value	Mean Delta
Baseline (Current)	0.7643	9.11×10^{-151}	+3.86
Halved Risk	0.7641	1.22×10^{-150}	+3.85
Doubled Risk	0.7651	3.02×10^{-151}	+3.85
Uniform Risk (all=2.0)	0.7637	2.12×10^{-150}	+3.81
No Protective	0.7636	2.40×10^{-150}	+3.78
Doubled Protective	0.7637	2.19×10^{-150}	+3.93
Random (seed=42)	0.7637	2.13×10^{-150}	+3.81

All seven scenarios produce virtually identical correlations with AIOE, ranging from 0.7636 to 0.7651, with every p -value below 10^{-150} . Whether risk multipliers are halved, doubled, set uniformly to 2.0, or replaced with random values, the relative ordering of occupations remains essentially unchanged. The occupation rankings are driven by the structural classification of features into risk and protective categories and by the binary count of features above median (the uplift formula), not by the specific numeric values of the multipliers. This converts what might initially appear as a weakness of the methodology (subjective multiplier choice) into a demonstrated strength (multiplier-invariant structural stability).

4.5 TREE SATURATION—AN ARCHITECTURAL ANALYSIS

During the stability testing phase, I discovered an important architectural property of the Stacked Ensemble that deserves careful discussion. When I compared the prediction deltas (difference between multiplied and original feature predictions) across different model architectures, the results showed a clear pattern:

The Stacked Ensemble produces deltas with a mean of just +0.14 and a range of essentially zero to 0.6. Compare this to Ridge Regression, which produces deltas spanning from

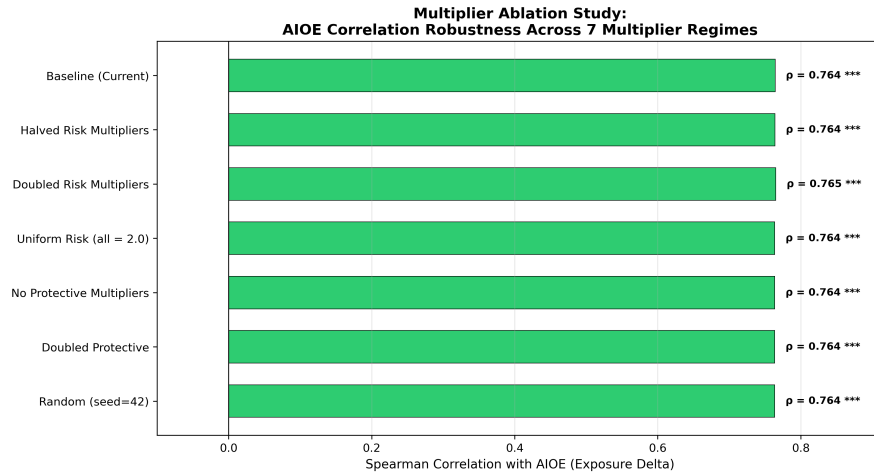


Figure 4.6: Multiplier Ablation Study. All seven scenarios produce AIOE correlations within the narrow range of 0.7636–0.7651, demonstrating structural invariance to multiplier specification.

Table 4.3: Tree Extrapolation Delta Statistics by Model Architecture

Model	Mean Delta	Std Dev	Range
Ridge	−30.95	42.33	[−203.2, +149.3]
XGBoost	+1.67	4.96	[−15.6, +14.9]
Random Forest	+0.13	4.73	[−24.8, +10.4]
Stacked Ensemble	+0.14	0.18	[−0.0, +0.6]
SVR (RBF)	+3.41	28.78	[−53.9, +50.1]

–203 to +149. XGBoost and Random Forest are tree-based models that partition the feature space into rectangular regions during training. When a feature value is multiplied beyond the range the model saw during training (for instance, a Programming score of 100 being multiplied to 350), the tree maps it to the nearest existing leaf node, producing nearly the same prediction as before. Trees cannot extrapolate by design.

The Stacked Ensemble, whose predictions are dominated by its tree-based base learners, inherits this saturation behavior. The meta-learner (Ridge) operates on the base learners' predictions, so if the base predictions do not change, neither does the meta-learner's output. Ridge Regression alone can extrapolate linearly, producing wild deltas when features are multiplied by 3.5×, but there is no guarantee that such extrapolation beyond the training distribution is meaningful.

This finding explains why the three-layer architecture works. Layer 1 (the ML model) validates that Fluency of Ideas, Originality, and other key features genuinely predict 2013 automation risk. It confirms the structural plausibility of the feature set. But Layer 1 does not produce the MAEI rankings. Layer 2 (feature classification into risk versus protective categories) and Layer 3 (the uplift formula, which counts how many risk and protective features each occupation has above the median) generate the actual occupation ordering. The uplift formula operates on simple feature counts: how many risk features does this occupation have? It operates on feature counts rather than model predictions, making it immune to the tree saturation limitation.

I treat this as a methodological feature rather than a flaw. The model validates the feature selection; the uplift formula produces the rankings. Each component has a clear, defensible role, and presenting them separately is more scientifically rigorous than claiming the entire pipeline is a monolithic “ML predicts automation” system.

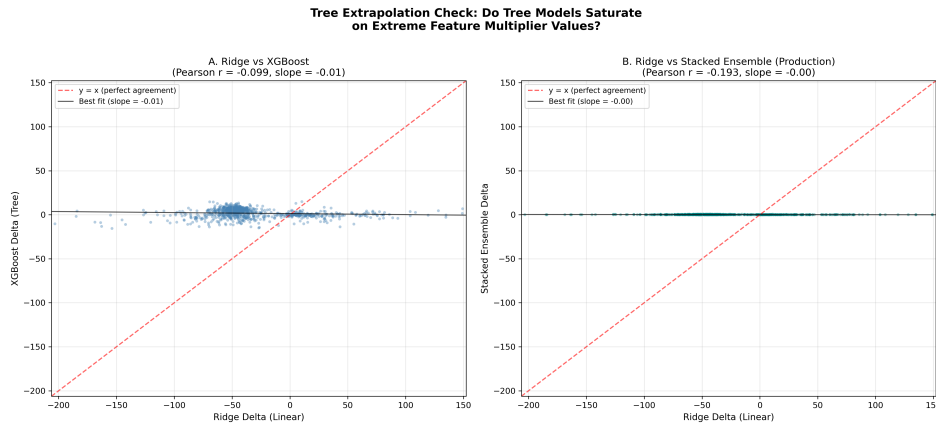


Figure 4.7: Tree Extrapolation Comparison. Ridge Regression shows extrapolation deltas spanning -203 to $+149$, while the Stacked Ensemble’s tree-based components produce near-zero deltas (range: -0.0 to $+0.6$), confirming the saturation hypothesis.

4.6 DISCUSSION

Collectively, these results support three conclusions. First, the occupational profile of AI exposure has structurally inverted since 2013. The jobs Frey and Osborne (2017) identified as safe (cognitive, analytical, communicative) are precisely the ones the MAEI identifies as most exposed under 2026 conditions. This finding aligns with Felten *et al.* (2021), Eloundou *et al.* (2023), and Anthropic’s (2026) independent analyses using different data and methods.

Second, the MAEI’s occupation rankings are structurally stable. The seven-scenario ablation study proves that the rankings are invariant to multiplier specification; regardless of whether multiplier values are halved, doubled, randomized, or set uniformly, the same occupations end up at the top and bottom of the exposure distribution. This stability stems from the feature classification layer (risk vs. protective) and the uplift formula, which depend on which features are classified as risk or protective and how many of them each occupation possesses, not on the specific multiplier magnitudes.

Third, the three-layer architecture demonstrates a methodologically explicit approach to scenario-based indexing. By explicitly documenting the tree saturation phenomenon

and separating the model's validation role from the uplift formula's ranking role, the MAEI avoids the common pitfall of over-claiming what the machine learning component actually contributes. This decomposition makes the index more interpretable and more defensible than a black-box approach would be.

CHAPTER 5

LIMITATIONS, FUTURE DIRECTIONS, AND CONCLUSION

5.1 LIMITATIONS

Every model has constraints, and they should be stated plainly.

Scenario Calibration vs. Empirical Estimation. The MAEI is a structured counterfactual scenario, not an econometric estimation trained on observed labor market displacement. Such data simply does not exist yet; federal employment statistics have not had time to capture the effects of technologies deployed only since 2023. Anthropic (2026) found limited employment effects to date, suggesting the full labor market impact of GenAI may take years to materialize. Until longitudinal displacement data becomes available, the MAEI remains a “what-if” structural model rather than a predictive one.

Circular Uplift Calibration. The Broad Exposure Uplift weights ($W_{\text{exposure}} = 10$, $W_{\text{protection}} = 3$) were calibrated to approximately match Eloundou *et al.* (2023)’s finding that 80% of the workforce faces some LLM exposure. The sensitivity analysis (Table 4.1) shows that all tested configurations produce qualitatively similar results (82.9%–87.5% of occupations affected), but the specific weight values are informed by the same literature that motivates the research question. This circularity does not invalidate the rankings (which are stable across weight changes) but limits the ability to treat the absolute exposure magnitudes as independently derived.

Tree Saturation. As documented in Section 4.5, the Stacked Ensemble’s tree-based components produce near-zero prediction deltas when features are multiplied beyond the training range. The ML model validates feature selection and structural plausibility but does not drive the final occupation rankings. Future work could explore whether linear or

kernel-based models that can extrapolate would produce more informative model-driven deltas, though the meaningfulness of extrapolating far beyond training distributions is itself debatable.

Shared O*NET Substrate. Both the MAEI and Felten *et al.* (2021)’s AIOE dataset are constructed from O*NET occupational data. While the measurement approaches differ substantially (scenario-based multipliers vs. crowdsourced AI-task linkages), the shared underlying data source means the $\rho = +0.764$ correlation may partially reflect the common O*NET structure rather than fully independent methodological convergence.

Ontological Ambiguity of “Exposure.” The MAEI measures capability overlap, not job loss. As Acemoglu and Restrepo (2018) emphasize, the path from task automation to job displacement runs through economic adoption costs, regulatory barriers, organizational inertia, and worker adaptability. A high MAEI score for “Lawyer” means AI can perform many legal tasks; it does not predict how many lawyers will be employed five years from now.

O*NET Data Latency. The O*NET database is updated on a rolling basis, and definitions for recently emerged occupations (for example, “Prompt Engineer” or “AI Safety Researcher”) may not yet be captured in the current taxonomy.

5.2 FUTURE DIRECTIONS

Three research extensions emerge from the MAEI framework:

Wage Shock Simulations. By joining the MAEI with Bureau of Labor Statistics Occupational Employment and Wage Statistics (OEWS), researchers can calculate the total wage volume “exposed” to AI capability overlap. Rather than predicting absolute job loss, future simulations could model wage-compression scenarios: if GenAI increases productivity in high-MAEI roles by 30%, how does the resulting supply shock of cognitive

output depress median wages in those sectors?

The “Human Premium.” Isolating occupations with high protective feature scores (Originality, Therapy and Counseling, Social Perceptiveness) and regressing these against longitudinal wage data could empirically test whether demand for distinctly human skills is increasing. If AI commoditizes cognitive output while human authenticity, empathy, and creative originality become scarcer, classic economic theory predicts that wages for these human-centric skills should rise, a measurable “Human Premium.”

Labor Reallocation Networks. Using cosine similarity between the O*NET feature vectors of high-MAEI and low-MAEI occupations, future work could construct reskilling pathway maps. These algorithmic “lifeboats” would identify the most efficient transitions for workers in highly exposed cognitive roles to move into roles with strong human structural advantages, minimizing the retraining distance while maximizing the reduction in AI exposure.

5.3 CONCLUSION

Revisiting the three research questions posed in Chapter 1, the evidence supports affirmative answers to each. For RQ1 (structural plausibility), the Stacked Ensemble’s cross-validated R^2 of 0.6794 and test R^2 of 0.6433 demonstrate that 2013 automation probabilities can be reliably predicted from modern O*NET features, and the feature importance hierarchy (Fluency of Ideas at 0.2177, Originality at 0.1919) confirms that the model learned Frey and Osborne (2017)’s theoretical bottleneck structure rather than statistical noise. For RQ2 (convergence with external benchmarks), the Spearman correlation of $\rho = +0.764$ ($p < 10^{-150}$) between the pure exposure delta and Felten *et al.* (2021)’s independently constructed AIOE provides strong evidence of construct validity. The dual-correlation signal ($\rho = -0.357$ anchored, $\rho = +0.764$ pure delta) is a quantitative marker of the shift in risk profile. For RQ3 (robustness to assumptions), the seven-scenario ablation study (ρ range: 0.7636–0.7651) shows that occupation

rankings are invariant to multiplier specification, converting a potential methodological vulnerability into a demonstrated structural strength.

The Modern AI Exposure Index bridges the analytical gap between the robotics-focused automation anxiety of the 2010s and the cognitive-AI reality of the 2020s. Through a five-stage computational pipeline (data engineering, NLP feature extraction, ensemble modeling, scenario-based perturbation, and external validation) I constructed a transparent, reproducible map of occupational exposure to Generative AI across 1,016 U.S. occupations.

The external validation produced a dual-correlation signal: $\rho = +0.764$ ($p < 10^{-150}$) for the pure exposure delta against Felten *et al.* (2021)'s AIOE benchmark, and $\rho = -0.357$ for the historically anchored score. Together, these correlations provide evidence of both construct validity and the structural inversion in automation risk: the same occupations rated as safe in 2013 are now rated as highly exposed to AI capability overlap.

The seven-scenario multiplier ablation study (ρ range: 0.7636–0.7651) proves that the occupation rankings are structurally stable, driven by the classification of features into risk and protective categories rather than by specific multiplier values. The explicit decomposition of the pipeline into a three-layer architecture (where the ML model validates feature selection, the feature classification encodes domain knowledge, and the uplift formula generates rankings) makes the MAEI more interpretable and defensible than a monolithic machine learning approach.

The MAEI does not claim to predict unemployment or forecast the economic trajectory of any specific occupation. What it provides is a rigorous, parameter-bounded framework for understanding which occupations overlap most with current AI capabilities, how that profile compares to the historical baseline, and how stable those conclusions are across a wide range of assumptions. As Generative AI continues to evolve, the MAEI's fully open-source pipeline can be recalibrated with updated multipliers, extended to

new O*NET releases, and validated against emerging labor market data as it becomes available.

In the broadest terms, this dissertation contributes to a growing body of evidence that the relationship between human labor and artificial intelligence is entering a fundamentally new phase. The occupations that built the middle class of the 20th century (manufacturing, construction, transportation) were the primary targets of the Second Wave of automation. The occupations that promised upward mobility in the 21st century (law, finance, programming, analysis) are now the primary targets of the Third Wave. The MAEI does not predict how this transition will unfold, but it provides a structural map that policymakers, educators, and workers can use to navigate it. Understanding which capabilities overlap most with AI is the essential first step toward building educational curricula, workforce development programs, and economic policies that prepare the labor force for a world where cognitive work is no longer exclusively human.

APPENDIX A

COMPLETE FEATURE MULTIPLIER TABLE

Table A.1: Literature-Calibrated Feature Multipliers (2013 → 2026) — Risk Features
(34 features, multipliers 1.60×–3.50×, 30th percentile gate)

Feature	Category	Multiplier	Citation
Skills_Programming	Coding	3.50×	Peng et al. (2023)
Skills_Writing	Language	3.00×	Noy & Zhang (2023)
Knowledge_Clerical	Admin	3.00×	Eloundou et al. (2023)
WA_Processing Information	Data	3.00×	Eloundou et al. (2023)
Abilities_Written Expression	Language	2.80×	OpenAI (2023)
WA_Analyzing Data	Data	2.80×	General
Abilities_Written Comprehension	Language	2.70×	Eloundou et al. (2023)
WA_Documenting Info	Data	2.70×	General
Skills_Reading Comprehension	Language	2.60×	Eloundou et al. (2023)
Abilities_Deductive Reasoning	Reasoning	2.50×	Choi et al. (2023)
Knowledge_Law and Government	Legal	2.50×	Choi et al. (2023)
Skills_Mathematics	Math	2.50×	OpenAI (2023)
Know_Computers & Electronics	Coding	2.50×	Webb (2020)
Abilities_Info Ordering	Reasoning	2.50×	General
Know_Econ & Accounting	Math	2.40×	Brynjolfsson et al. (2023)
Abilities_Math Reasoning	Math	2.40×	OpenAI (2023)
Abilities_Inductive Reasoning	Reasoning	2.40×	General
WA_Getting Information	Admin	2.40×	General
WA_Scheduling Work	Admin	2.40×	General
Abilities_Number Facility	Math	2.30×	General
Knowledge_Mathematics	Math	2.30×	General
Abilities_Speech Recognition	Speech	2.30×	OpenAI (2023)
WA_Evaluating Compliance	Admin	2.30×	General
Know_Admin & Management	Admin	2.20×	General
Skills_Technology Design	Coding	2.20×	General
Skills_Active Listening	Speech	2.10×	General
Skills_Monitoring	Speech	2.10×	General
Skills_Complex Problem Solving	Reasoning	2.00×	General
Abilities_Oral Comprehension	Speech	2.00×	General
WA_Updating Knowledge	Admin	2.00×	General
WA_Interacting w/ Computers	Admin	2.00×	General
KW_routine_keywords	NLP	1.80×	Acemoglu & Restrepo (2018)
KW_technical_keywords	NLP	1.60×	Webb (2020)

Table A.2: Literature-Calibrated Feature Multipliers (2013 → 2026) — Protective Features (34 features, multipliers 1.02×–1.10×, 85th percentile gate)

Feature	Category	Multiplier	Citation
Know_Therapy & Counseling	Social	1.10×	General
Skills_Negotiation	Social	1.08×	Frey & Osborne (2017)
Knowledge_Psychology	Social	1.08×	General
WA_Establishing Relationships	Social	1.08×	General
Skills_Persuasion	Social	1.07×	General
WA_Resolving Conflicts	Social	1.07×	General
WA_Guiding Subordinates	Social	1.07×	General
WA_Thinking Creatively	Creative	1.06×	Frey & Osborne (2017)
Skills_Social Perceptiveness	Social	1.06×	Frey & Osborne (2017)
Skills_Service Orientation	Social	1.06×	General
WA_Working with Public	Social	1.06×	General
WA_Coaching Others	Social	1.06×	General
Interest_Social	Interest	1.06×	General
Abilities_Originality	Creative	1.05×	Doshi & Hauser (2024)
Abilities_Spatial Orientation	Physical	1.05×	Frey & Osborne (2017)
Abilities_Manual Dexterity	Physical	1.05×	IFR (2024)
WA_Operating Vehicles	Physical	1.05×	General
Skills_Instructing	Social	1.05×	General
WA_Training Others	Social	1.05×	General
Interest_Artistic	Interest	1.05×	General
Know_Sociology & Anthro	Social	1.05×	General
Abilities_Fluency of Ideas	Creative	1.04×	Doshi & Hauser (2024)
Skills_Coordination	Social	1.04×	General
Abilities_Finger Dexterity	Physical	1.04×	Frey & Osborne (2017)
WA_Physical Activities	Physical	1.04×	General
KW_physical keywords	NLP	1.04×	General
Abilities_Arm-Hand Steadiness	Physical	1.03×	General
Abilities_Control Precision	Physical	1.03×	General
Abilities_Depth Perception	Physical	1.03×	General
WA_Handling Objects	Physical	1.03×	General
KW_creative keywords	NLP	1.03×	General
Abilities_Far Vision	Physical	1.02×	General

APPENDIX B

COMPLETE MAEI OCCUPATION INDEX (923 OCCUPATIONS)

Table B.1: Complete MAEI Occupation Index — all 923 scored occupations sorted by MAEI 2026 Score (descending).

#	Occupation	MAEI 2026	Delta	Risk Tier
1	Photographic Process Workers and Processing Machine Operators	100.0	+2.24	Very High
2	Office Clerks, General	100.0	+4.31	Very High
3	Data Entry Keyers	100.0	+3.97	Very High
4	Secretaries and Administrative Assistants, Except Legal, Medical, and Executive	100.0	+5.04	Very High
5	Legal Secretaries and Administrative Assistants	100.0	+4.73	Very High
6	Claims Adjusters, Examiners, and Investigators	100.0	+8.07	Very High
7	Insurance Appraisers, Auto Damage	100.0	+4.49	Very High
8	Nuclear Power Reactor Operators	100.0	+6.08	Very High
9	Dispatchers, Except Police, Fire, and Ambulance	100.0	+5.36	Very High
10	Freight Forwarders	100.0	+3.83	Very High
11	Cargo and Freight Agents	100.0	+2.60	Very High
12	Receptionists and Information Clerks	100.0	+4.15	Very High
13	Order Clerks	100.0	+4.17	Very High
14	New Accounts Clerks	100.0	+3.86	Very High
15	File Clerks	100.0	+4.45	Very High

Continued on next page

Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
16	Credit Authorizers, Checkers, and Clerks	100.0	+3.25	Very High
17	Accountants and Auditors	100.0	+7.68	Very High
18	Brokerage Clerks	100.0	+5.60	Very High
19	Budget Analysts	100.0	+7.24	Very High
20	Credit Analysts	100.0	+5.89	Very High
21	Procurement Clerks	100.0	+6.08	Very High
22	Payroll and Timekeeping Clerks	100.0	+6.32	Very High
23	Insurance Underwriters	100.0	+4.44	Very High
24	Bookkeeping, Accounting, and Auditing Clerks	100.0	+4.72	Very High
25	Billing and Posting Clerks	100.0	+5.31	Very High
26	Loan Officers	100.0	+7.39	Very High
27	Tax Examiners and Collectors, and Revenue Agents	100.0	+7.22	Very High
28	Tax Preparers	100.0	+5.17	Very High
29	Telemarketers	100.0	+4.07	Very High
30	Real Estate Brokers	100.0	+5.97	Very High
31	Parts Salespersons	100.0	+5.04	Very High
32	Insurance Claims and Policy Processing Clerks	100.0	+2.97	Very High
33	Surveying and Mapping Technicians	100.0	+4.74	Very High
34	Paralegals and Legal Assistants	100.0	+6.05	Very High
35	Title Examiners, Abstractors, and Searchers	100.0	+5.01	Very High
36	Compensation and Benefits Managers	100.0	+6.47	Very High
37	Library Technicians	100.0	+1.37	Very High
38	Tellers	100.0	+1.96	Very High

Continued on next page

Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
39	Switchboard Operators, Including Answering Service	99.8	+3.84	Very High
40	Counter and Rental Clerks	99.8	+2.82	Very High
41	Shipping, Receiving, and Inventory Clerks	99.8	+1.76	Very High
42	Telephone Operators	99.5	+2.53	Very High
43	Milling and Planing Machine Setters, Operators, and Tenders, Metal and Plastic	99.5	+1.50	Very High
44	Inspectors, Testers, Sorters, Samplers, and Weighers	99.5	+1.50	Very High
45	Interviewers, Except Eligibility and Loan	99.4	+5.43	Very High
46	Camera and Photographic Equipment Repairers	99.4	+2.39	Very High
47	Prepress Technicians and Workers	99.2	+2.23	Very High
48	Locomotive Engineers	99.1	+3.12	Very High
49	Bill and Account Collectors	98.8	+3.75	Very High
50	Loan Interviewers and Clerks	98.5	+6.51	Very High
51	Woodworking Machine Setters, Operators, and Tenders, Except Sawing	98.5	+1.50	Very High
52	Textile Bleaching and Dyeing Machine Operators and Tenders	98.5	+1.50	Very High
53	Models	98.2	+0.19	Very High
54	Sewers, Hand	98.1	-0.86	Very High
55	Gambling Surveillance Officers and Gambling Investigators	98.1	+3.12	Very High
56	Watch and Clock Repairers	98.0	-0.95	Very High

Continued on next page

Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
57	Packaging and Filling Machine Operators and Tenders	98.0	+0.04	Very High
58	Motion Picture Projectionists	98.0	+0.95	Very High
59	Umpires, Referees, and Other Sports Officials	97.9	-0.11	Very High
60	Agricultural Inspectors	97.9	+3.87	Very High
61	Medical and Clinical Laboratory Technologists	97.8	+7.84	Very High
62	Team Assemblers	97.8	+0.80	Very High
63	Gambling Dealers	97.8	+1.78	Very High
64	Driver/Sales Workers	97.7	-0.35	Very High
65	Pharmacy Technicians	97.6	+5.63	Very High
66	Bridge and Lock Tenders	97.6	+0.62	Very High
67	Grinding and Polishing Workers, Hand	97.6	+0.61	Very High
68	Etchers and Engravers	97.5	-0.51	Very High
69	Ophthalmic Laboratory Technicians	97.5	+0.48	Very High
70	Log Graders and Scalers	97.5	+0.48	Very High
71	Aviation Inspectors	97.4	+7.41	Very High
72	Textile Winding, Twisting, and Drawing Out Machine Setters, Operators, and Tenders	97.2	+1.20	Very High
73	Cashiers	97.2	+0.17	Very High
74	Postal Service Clerks	97.1	+2.08	Very High
75	Ushers, Lobby Attendants, and Ticket Takers	97.1	+1.06	Very High
76	Crushing, Grinding, and Polishing Machine Setters, Operators, and Tenders	97.0	+0.03	Very High
77	Weighers, Measurers, Checkers, and Samplers, Recordkeeping	97.0	+1.95	Very High
78	Library Assistants, Clerical	96.9	+1.94	Very High

Continued on next page

Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
79	Model Makers, Wood	96.8	+0.76	Very High
80	Electromechanical Equipment Assemblers	96.6	-0.36	Very High
81	Cooks, Restaurant	96.6	+0.62	Very High
82	Shoe Machine Operators and Tenders	96.6	-0.39	Very High
83	Textile Cutting Machine Setters, Operators, and Tenders	96.5	+1.50	Very High
84	Cytogenetic Technologists	96.4	+6.38	Very High
85	Hosts and Hostesses, Restaurant, Lounge, and Coffee Shop	96.2	-0.78	Very High
86	Dental Laboratory Technicians	96.2	-0.84	Very High
87	Cytotechnologists	96.1	+6.06	Very High
88	Farm Labor Contractors	96.0	-1.00	Very High
89	Grinding, Lapping, Polishing, and Buffing Machine Tool Setters, Operators, and Tenders, Metal and Plastic	95.9	+0.91	Very High
90	Operating Engineers and Other Construction Equipment Operators	95.9	+0.90	Very High
91	Insurance Sales Agents	95.9	+3.86	Very High
92	Door-to-Door Sales Workers, News and Street Vendors, and Related Workers	95.8	+1.79	Very High
93	Gem and Diamond Workers	95.8	+0.76	Very High
94	Transportation Inspectors	95.8	+5.75	Very High
95	Pesticide Handlers, Sprayers, and Applicators, Vegetation	95.6	-1.43	Very High
96	Animal Breeders	95.5	+0.47	Very High
97	Hotel, Motel, and Resort Desk Clerks	95.2	+1.24	Very High

Continued on next page

Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
98	Electrical and Electronic Equipment Assemblers	95.2	+0.23	Very High
99	Landscaping and Groundskeeping Workers	95.2	+0.18	Very High
100	Molding, Coremaking, and Casting Machine Setters, Operators, and Tenders, Metal and Plastic	95.2	+0.18	Very High
101	Coin, Vending, and Amusement Machine Servicers and Repairers	95.1	+1.11	Very High
102	Print Binding and Finishing Workers	95.1	+0.09	Very High
103	Extruding, Forming, Pressing, and Compacting Machine Setters, Operators, and Tenders	95.1	+2.08	Very High
104	Manicurists and Pedicurists	95.0	+0.04	Very High
105	Adhesive Bonding Machine Operators and Tenders	95.0	+0.03	Very High
106	Rock Splitters, Quarry	95.0	-1.00	Very High
107	Retail Salespersons	95.0	+3.00	Very High
108	Technical Writers	94.9	+5.91	Very High
109	Human Resources Assistants, Except Payroll and Timekeeping	94.9	+4.88	Very High
110	Model Makers, Metal and Plastic	94.8	+1.80	Very High
111	First-Line Supervisors of Housekeeping and Janitorial Workers	94.8	+0.78	Very High
112	Cement Masons and Concrete Finishers	94.8	+0.76	Very High
113	Jewelers and Precious Stone and Metal Workers	94.6	-0.40	Very High

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
114	Welders, Cutters, Solderers, and Brazers	94.5	+0.47	Very High
115	Radio, Cellular, and Tower Equipment Installers and Repairers	94.4	+1.43	Very High
116	Conveyor Operators and Tenders	94.4	+1.36	Very High
117	Waiters and Waitresses	94.3	+0.35	Very High
118	Traffic Technicians	94.3	+4.31	Very High
119	Cooling and Freezing Equipment Operators and Tenders	94.2	+1.22	Very High
120	Bicycle Repairers	94.2	+0.19	Very High
121	Couriers and Messengers	94.2	+0.18	Very High
122	Service Unit Operators, Oil and Gas	94.1	+1.08	Very High
123	Outdoor Power Equipment and Other Small Engine Mechanics	94.1	+1.07	Very High
124	Cartographers and Photogrammetrists	94.1	+6.07	Very High
125	Rail Yard Engineers, Dinkey Operators, and Hostlers	94.0	+2.97	Very High
126	Office Machine Operators, Except Computer	93.9	+1.94	Very High
127	Mail Clerks and Mail Machine Operators, Except Postal Service	93.9	-0.11	Very High
128	Histotechnologists	93.7	+3.71	Very High
129	Cabinetmakers and Bench Carpenters	93.7	+1.66	Very High
130	Helpers–Painters, Paperhangers, Plasterers, and Stucco Masons	93.6	-0.36	Very High
131	Tire Builders	93.6	-0.38	Very High
132	Machine Feeders and Offbearers	93.6	+0.61	Very High

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
133	Drilling and Boring Machine Tool Setters, Operators, and Tenders, Metal and Plastic	93.5	-0.51	Very High
134	Production, Planning, and Expediting Clerks	93.5	+5.47	Very High
135	Meat, Poultry, and Fish Cutters and Trimmers	93.3	-0.65	Very High
136	Patternmakers, Wood	93.2	+2.24	Very High
137	Cooks, Short Order	93.0	-1.00	Very High
138	Medical Transcriptionists	93.0	+4.00	Very High
139	Forging Machine Setters, Operators, and Tenders, Metal and Plastic	92.9	-0.06	Very High
140	Refuse and Recyclable Material Collectors	92.9	-0.08	Very High
141	Industrial Truck and Tractor Operators	92.9	-0.11	Very High
142	Food and Tobacco Roasting, Baking, and Drying Machine Operators and Tenders	92.8	+1.79	Very High
143	Gambling and Sports Book Writers and Runners	92.6	+1.64	Very High
144	Glass Blowers, Molders, Benders, and Finishers	92.5	+2.52	Very High
145	Heat Treating Equipment Setters, Operators, and Tenders, Metal and Plastic	92.5	+1.49	Very High
146	Nuclear Monitoring Technicians	92.4	+7.39	Very High
147	Butchers and Meat Cutters	92.4	-0.64	Very High
148	Multiple Machine Tool Setters, Operators, and Tenders, Metal and Plastic	92.2	+1.23	Very High
149	Dredge Operators	92.2	+0.22	Very High
150	Executive Secretaries and Executive Administrative Assistants	92.2	+6.16	Very High

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
151	Gas Compressor and Gas Pumping Station Operators	92.0	+0.97	Very High
152	Electrical and Electronics Installers and Repairers, Transportation Equipment	91.8	+0.83	Very High
153	Pump Operators, Except Wellhead Pumpers	91.7	+1.66	Very High
154	Plating Machine Setters, Operators, and Tenders, Metal and Plastic	91.6	-0.42	Very High
155	Crane and Tower Operators	91.5	+1.51	Very High
156	Signal and Track Switch Repairers	91.4	+1.38	Very High
157	Patternmakers, Metal and Plastic	91.4	+1.37	Very High
158	Extruding and Drawing Machine Setters, Operators, and Tenders, Metal and Plastic	91.4	+0.37	Very High
159	Tour Guides and Escorts	91.3	+0.35	Very High
160	Painting, Coating, and Decorating Workers	91.3	-0.67	Very High
161	Musical Instrument Repairers and Tuners	91.3	+0.30	Very High
162	Stationary Engineers and Boiler Operators	91.2	+2.24	Very High
163	Nuclear Technicians	91.2	+6.22	Very High
164	Helpers–Carpenters	91.2	-0.79	Very High
165	Transportation Vehicle, Equipment and Systems Inspectors, Except Aviation	91.2	+1.19	Very High
166	Fence Erectors	90.9	-1.10	Very High
167	Buyers and Purchasing Agents, Farm Products	90.7	+3.72	Very High
168	Semiconductor Processing Technicians	90.7	+2.67	Very High
169	Automotive Body and Related Repairers	90.5	-0.52	Very High
170	Tool Grinders, Filers, and Sharpeners	90.1	+2.09	Very High

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
171	Electrical and Electronic Engineering Technologists and Technicians	90.1	+6.08	Very High
172	Mechanical Door Repairers	90.0	-1.00	Very High
173	Stone Cutters and Carvers, Manufacturing	89.9	-0.09	Very High
174	Biofuels Processing Technicians	89.7	+3.72	Very High
175	Separating, Filtering, Clarifying, Precipitating, and Still Machine Setters, Operators, and Tenders	89.7	+1.65	Very High
176	Bakers	89.6	+0.62	Very High
177	Sewing Machine Operators	89.5	+0.46	Very High
178	Dining Room and Cafeteria Attendants and Bartender Helpers	89.4	-1.57	Very High
179	Construction Laborers	89.4	+1.37	Very High
180	Real Estate Sales Agents	89.3	+3.35	Very High
181	Molders, Shapers, and Casters, Except Metal and Plastic	89.3	-0.67	Very High
182	Potters, Manufacturing	89.3	-0.68	Very High
183	Hydroelectric Plant Technicians	89.3	+4.31	Very High
184	Correspondence Clerks	89.3	+3.27	Very High
185	Sales Representatives, Wholesale and Manufacturing, Except Technical and Scientific Products	89.2	+4.20	Very High
186	Reinforcing Iron and Rebar Workers	89.1	-0.86	Very High
187	Roofers	89.1	-0.87	Very High
188	Metal-Refining Furnace Operators and Tenders	89.0	+0.95	Very High

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
189	Rail-Track Laying and Maintenance Equipment Operators	88.9	-0.07	Very High
190	Extruding and Forming Machine Setters, Operators, and Tenders, Synthetic and Glass Fibers	88.9	+0.92	Very High
191	Stonemasons	88.9	-0.12	Very High
192	Chemical Plant and System Operators	88.8	+3.85	Very High
193	Riggers	88.5	-0.55	Very High
194	Rail Car Repairers	88.4	+0.36	Very High
195	Power Plant Operators	88.1	+3.12	Very High
196	Proofreaders and Copy Markers	87.8	+3.83	Very High
197	Property, Real Estate, and Community Association Managers	87.8	+6.81	Very High
198	Cutting and Slicing Machine Setters, Operators, and Tenders	87.8	+1.80	Very High
199	Robotics Technicians	87.0	+5.95	Very High
200	Terrazzo Workers and Finishers	86.9	-1.11	Very High
201	Forest and Conservation Workers	86.9	-0.11	Very High
202	Carpet Installers	86.8	-0.22	Very High
203	Sawing Machine Setters, Operators, and Tenders, Wood	86.6	+0.61	Very High
204	Electrical and Electronics Drafters	86.6	+5.61	Very High
205	Biomass Plant Technicians	86.5	+1.51	Very High
206	Meter Readers, Utilities	86.3	+1.35	Very High
207	Parking Attendants	86.2	-0.79	Very High

Continued on next page

Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
208	Veterinary Assistants and Laboratory Animal Caretakers	86.2	+0.21	Very High
209	Paperhangers	86.2	-0.80	Very High
210	Subway and Streetcar Operators	86.2	+0.20	Very High
211	Food Servers, Nonrestaurant	86.2	+0.19	Very High
212	Pourers and Casters, Metal	86.1	-0.86	Very High
213	Maintenance Workers, Machinery	86.1	+0.08	Very High
214	Medical Secretaries and Administrative Assistants	86.0	+5.04	Very High
215	Laborers and Freight, Stock, and Material Movers, Hand	85.9	+0.93	Very High
216	Food Preparation Workers	85.9	-1.09	Very High
217	Floor Sanders and Finishers	85.9	-1.14	Very High
218	Layout Workers, Metal and Plastic	85.7	+1.65	Very High
219	Highway Maintenance Workers	85.6	-1.38	Very High
220	Tool and Die Makers	85.5	+1.51	Very High
221	Security Guards	85.5	+1.51	Very High
222	Furniture Finishers	85.5	-1.52	Very High
223	Railroad Conductors and Yardmasters	85.4	+2.40	Very High
224	Electro-Mechanical and Mechatronics Technologists and Technicians	85.3	+4.31	Very High
225	Parking Enforcement Workers	85.2	+1.22	Very High
226	Recycling and Reclamation Workers	85.2	+0.17	Very High
227	Word Processors and Typists	84.7	+3.68	Very High
228	Baggage Porters and Bellhops	84.7	+1.67	Very High
229	Printing Press Operators	84.5	+1.51	Very High

Continued on next page

Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
230	Automotive and Watercraft Service Attendants	84.2	+1.22	Very High
231	Purchasing Agents, Except Wholesale, Retail, and Farm Products	84.1	+7.12	Very High
232	Lathe and Turning Machine Tool Setters, Operators, and Tenders, Metal and Plastic	84.1	+0.07	Very High
233	Security and Fire Alarm Systems Installers	83.9	+1.94	Very High
234	Wellhead Pumpers	83.8	-0.22	Very High
235	Sailors and Marine Oilers	83.6	+0.63	Very High
236	Paving, Surfacing, and Tamping Equipment Operators	83.6	+0.62	Very High
237	Sheet Metal Workers	83.5	+1.51	Very High
238	Rolling Machine Setters, Operators, and Tenders, Metal and Plastic	83.5	+0.48	Very High
239	Septic Tank Servicers and Sewer Pipe Cleaners	83.4	+0.36	Very High
240	Cooks, Institution and Cafeteria	83.3	+0.35	Very High
241	Tailors, Dressmakers, and Custom Sewers	83.3	-0.69	Very High
242	Gambling Change Persons and Booth Cashiers	83.2	+0.19	Very High
243	Plasterers and Stucco Masons	83.1	-0.86	Very High
244	Mixing and Blending Machine Setters, Operators, and Tenders	83.0	+0.04	Very High
245	Helpers—Brickmasons, Blockmasons, Stonemasons, and Tile and Marble Setters	82.8	-0.23	Very High
246	Insulation Workers, Floor, Ceiling, and Wall	82.3	-0.71	Very High
247	Engine and Other Machine Assemblers	82.2	+0.23	Very High
248	Cleaning, Washing, and Metal Pickling Equipment Operators and Tenders	82.2	+1.22	Very High

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
249	Pile Driver Operators	82.1	+0.08	Very High
250	Structural Iron and Steel Workers	82.0	-0.96	Very High
251	Civil Engineering Technologists and Technicians	82.0	+7.01	Very High
252	Animal Caretakers	81.9	-0.06	Very High
253	Archivists	81.9	+5.94	Very High
254	Cooks, Fast Food	81.8	+0.76	Very High
255	Brickmasons and Blockmasons	81.7	-0.27	Very High
256	Segmental Pavers	81.7	-1.35	Very High
257	Refractory Materials Repairers, Except Brickmasons	81.3	-0.71	Very High
258	Gas Plant Operators	81.3	+3.28	Very High
259	Pressers, Textile, Garment, and Related Materials	81.2	+0.17	Very High
260	Motorcycle Mechanics	80.9	+1.86	Very High
261	Administrative Services Managers	80.7	+7.71	Very High
262	Postmasters and Mail Superintendents	80.6	+5.56	Very High
263	Derrick Operators, Oil and Gas	80.4	+0.39	Very High
264	Heavy and Tractor-Trailer Truck Drivers	80.2	+1.21	Very High
265	Postal Service Mail Sorters, Processors, and Processing Machine Operators	80.2	+1.20	Very High
266	Medical Equipment Preparers	80.0	+1.96	High
267	Barbers	79.9	-0.11	High
268	Aircraft Structure, Surfaces, Rigging, and Systems Assemblers	79.8	+0.76	High

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
269	Helpers–Installation, Maintenance, and Repair Workers	79.5	+0.53	High
270	Logging Equipment Operators	79.3	+0.34	High
271	Chemical Equipment Operators and Tenders	79.3	+3.27	High
272	Shampooers	78.8	-0.20	High
273	Locksmiths and Safe Repairers	78.4	+1.40	High
274	Cutting, Punching, and Press Machine Setters, Operators, and Tenders, Metal and Plastic	78.3	+0.31	High
275	Drywall and Ceiling Tile Installers	78.3	-0.73	High
276	Electric Motor, Power Tool, and Related Repairers	78.3	+2.25	High
277	Farm Equipment Mechanics and Service Technicians	78.0	+3.03	High
278	Floor Layers, Except Carpet, Wood, and Hard Tiles	78.0	-1.00	High
279	Eligibility Interviewers, Government Programs	77.7	+7.69	High
280	Broadcast Technicians	77.6	+3.58	High
281	Airfield Operations Specialists	77.0	+5.95	High
282	Computer, Automated Teller, and Office Machine Repairers	76.7	+2.69	High
283	Dishwashers	76.6	-0.36	High
284	Tree Trimmers and Pruners	76.3	-0.70	High
285	Passenger Attendants	76.2	+1.23	High
286	Bartenders	76.2	-0.80	High
287	Bus and Truck Mechanics and Diesel Engine Specialists	76.1	+3.12	High

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
288	Fallers	76.0	+0.01	High
289	Aircraft Mechanics and Service Technicians	75.9	+4.90	High
290	Opticians, Dispensing	75.8	+4.78	High
291	Pharmacy Aides	75.4	+3.41	High
292	Coil Winders, Tapers, and Finishers	75.2	+2.23	High
293	Home Appliance Repairers	74.5	+2.54	High
294	Helpers–Electricians	74.5	+0.47	High
295	Tile and Stone Setters	74.5	-0.55	High
296	Petroleum Pump System Operators, Refinery Operators, and Gaugers	74.4	+3.40	High
297	Atmospheric and Space Scientists	74.3	+7.28	High
298	Textile Knitting and Weaving Machine Setters, Operators, and Tenders	74.2	+1.21	High
299	Painters, Construction and Maintenance	73.9	-1.14	High
300	Statistical Assistants	73.8	+7.82	High
301	Avionics Technicians	73.8	+3.77	High
302	Amusement and Recreation Attendants	73.1	+1.08	High
303	Tank Car, Truck, and Ship Loaders	72.9	+0.92	High
304	Carpenters	72.8	+0.77	High
305	Glaziers	72.2	-0.85	High
306	Helpers–Roofers	71.7	-0.35	High
307	Power Distributors and Dispatchers	71.2	+7.22	High
308	Social Science Research Assistants	71.1	+6.06	High
309	Construction and Building Inspectors	71.0	+7.95	High
310	Energy Auditors	70.7	+7.69	High
311	Light Truck Drivers	70.7	+1.65	High

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
312	Tire Repairers and Changers	70.6	+0.61	High
313	Laundry and Dry-Cleaning Workers	70.4	-0.57	High
314	Mechanical Drafters	70.4	+2.42	High
315	Pest Control Workers	70.4	+4.41	High
316	Administrative Law Judges, Adjudicators, and Hearing Officers	70.2	+6.18	High
317	Food Batchmakers	69.9	-0.11	High
318	Maids and Housekeeping Cleaners	69.3	+0.32	High
319	Geoscientists, Except Hydrologists and Geographers	68.7	+5.70	High
320	Motorboat Mechanics and Service Technicians	68.6	+2.59	High
321	Industrial Machinery Mechanics	68.5	+1.51	High
322	Postal Service Mail Carriers	68.5	+0.47	High
323	Boilermakers	68.4	+0.36	High
324	Audiovisual Equipment Installers and Repairers	68.1	+3.13	High
325	Quality Control Analysts	68.1	+7.09	High
326	Lifeguards, Ski Patrol, and Other Recreational Protective Service Workers	68.1	+1.07	High
327	Search Marketing Strategists	68.0	+7.00	High
328	Hoist and Winch Operators	67.7	+2.70	High
329	Remote Sensing Technicians	67.7	+6.65	High
330	Heating, Air Conditioning, and Refrigeration Mechanics and Installers	67.4	+2.40	High
331	Paper Goods Machine Setters, Operators, and Tenders	67.1	+0.08	High

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
332	Roustabouts, Oil and Gas	67.0	-1.00	High
333	Foundry Mold and Coremakers	66.7	-0.28	High
334	Machinists	66.7	+1.66	High
335	Market Research Analysts and Marketing Specialists	66.6	+5.64	High
336	Maintenance and Repair Workers, General	66.2	+2.24	High
337	First-Line Supervisors of Food Preparation and Serving Workers	66.1	+3.09	High
338	Helpers—Production Workers	65.9	-0.10	High
339	Transportation, Storage, and Distribution Managers	65.1	+6.09	High
340	Janitors and Cleaners, Except Maids and Housekeeping Cleaners	65.0	-0.95	High
341	Supply Chain Managers	64.6	+5.64	High
342	Correctional Officers and Jailers	64.4	+4.42	High
343	Water and Wastewater Treatment Plant and System Operators	64.4	+3.40	High
344	Insulation Workers, Mechanical	64.2	+0.17	High
345	Control and Valve Installers and Repairers, Except Mechanical Door	63.9	+0.91	High
346	Automotive Service Technicians and Mechanics	63.9	+4.88	High
347	Cost Estimators	63.6	+6.65	High
348	Reservation and Transportation Ticket Agents and Travel Clerks	63.6	+2.56	High
349	Chemical Technicians	63.4	+6.37	High

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
350	Cutters and Trimmers, Hand	63.3	-0.71	High
351	Electronic Equipment Installers and Repairers, Motor Vehicles	63.3	+2.29	High
352	Personal Financial Advisors	63.0	+4.98	High
353	Transit and Railroad Police	62.7	+5.74	High
354	Pipelayers	62.1	+0.07	High
355	Museum Technicians and Conservators	61.9	+2.86	High
356	Motorboat Operators	61.8	-0.24	High
357	Commercial Pilots	61.7	+6.69	High
358	Physical Therapist Aides	61.7	+0.67	High
359	Welding, Soldering, and Brazing Machine Setters, Operators, and Tenders	61.6	+0.62	High
360	Tapers	61.3	-0.67	High
361	Camera Operators, Television, Video, and Film	61.1	+1.15	High
362	Food Cooking Machine Operators and Tenders	60.8	-0.25	High
363	Slaughterers and Meat Packers	60.5	+0.46	High
364	Millwrights	60.2	+1.24	High
365	Customer Service Representatives	60.2	+5.16	High
366	Costume Attendants	59.5	-1.54	Medium
367	First-Line Supervisors of Farming, Fishing, and Forestry Workers	59.2	+2.17	Medium
368	First-Line Supervisors of Landscaping, Lawn Service, and Groundskeeping Workers	59.1	+2.13	Medium
369	Advertising Sales Agents	59.0	+5.05	Medium
370	Recreational Vehicle Service Technicians	58.6	-0.41	Medium
371	Audio and Video Technicians	57.9	+2.87	Medium

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
372	Architectural and Civil Drafters	56.6	+4.61	Medium
373	Helpers–Pipelayers, Plumbers, Pipefitters, and Steamfitters	56.6	-0.42	Medium
374	Agricultural Engineers	56.4	+7.42	Medium
375	Hazardous Materials Removal Workers	55.5	+2.54	Medium
376	Public Safety Telecommunicators	55.5	+6.51	Medium
377	Embalmers	55.1	+1.08	Medium
378	Aerospace Engineering and Operations Technologists and Technicians	55.0	+6.96	Medium
379	Automotive Glass Installers and Repairers	54.3	-0.68	Medium
380	Rotary Drill Operators, Oil and Gas	54.3	+1.27	Medium
381	Continuous Mining Machine Operators	54.1	+0.07	Medium
382	Fire Inspectors and Investigators	53.1	+5.06	Medium
383	Dental Assistants	52.7	+1.72	Medium
384	Massage Therapists	52.5	-1.52	Medium
385	Telecommunications Line Installers and Repairers	52.0	+2.97	Medium
386	Compensation, Benefits, and Job Analysis Specialists	51.9	+4.93	Medium
387	Remote Sensing Scientists and Technologists	51.9	+8.86	Medium
388	Demonstrators and Product Promoters	51.3	+0.25	Medium
389	Shoe and Leather Workers and Repairers	51.0	-0.95	Medium
390	Histology Technicians	50.1	+3.09	Medium
391	Medical and Clinical Laboratory Technicians	50.0	+2.99	Medium
392	Economists	50.0	+6.96	Medium
393	Environmental Economists	49.1	+6.13	Medium

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
394	Court, Municipal, and License Clerks	49.1	+3.11	Medium
395	Historians	48.4	+4.44	Medium
396	Medical Appliance Technicians	48.1	+3.14	Medium
397	Crossing Guards and Flaggers	47.8	-1.23	Medium
398	Merchandise Displayers and Window Trimmers	46.8	-1.20	Medium
399	Judicial Law Clerks	46.2	+5.18	Medium
400	Geodetic Surveyors	46.1	+8.13	Medium
401	Judges, Magistrate Judges, and Magistrates	44.7	+4.70	Medium
402	Mechanical Engineering Technologists and Technicians	44.3	+6.25	Medium
403	Automotive Engineering Technicians	44.1	+6.07	Medium
404	Electrical and Electronics Repairers, Commercial and Industrial Equipment	43.6	+2.56	Medium
405	Locker Room, Coatroom, and Dressing Room Attendants	43.5	+0.52	Medium
406	Surveyors	43.5	+5.50	Medium
407	Interpreters and Translators	43.3	+5.34	Medium
408	Elevator and Escalator Installers and Repairers	42.9	+3.85	Medium
409	Mobile Heavy Equipment Mechanics, Except Engines	42.4	+2.41	Medium
410	Intelligence Analysts	42.3	+8.26	Medium
411	Structural Metal Fabricators and Fitters	41.9	+0.89	Medium
412	Graders and Sorters, Agricultural Products	41.9	+0.89	Medium
413	Diagnostic Medical Sonographers	41.4	+6.38	Medium
414	Gambling Cage Workers	40.6	+1.64	Medium

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
415	Detectives and Criminal Investigators	40.5	+6.51	Medium
416	Telecommunications Equipment Installers and Repairers, Except Line Installers	40.5	+4.47	Medium
417	Electrical and Electronics Repairers, Powerhouse, Substation, and Relay	40.4	+2.44	Medium
418	Police Identification and Records Officers	40.0	+5.95	Low
419	Radiation Therapists	39.9	+5.94	Low
420	Fraud Examiners, Investigators and Analysts	39.8	+6.80	Low
421	Financial Quantitative Analysts	39.3	+6.29	Low
422	Actors	39.0	+1.98	Low
423	Upholsterers	38.7	-0.34	Low
424	Private Detectives and Investigators	38.5	+7.55	Low
425	Biological Technicians	38.1	+8.14	Low
426	Bailiffs	38.1	+2.10	Low
427	Packers and Packagers, Hand	38.0	+0.03	Low
428	Solar Thermal Installers and Technicians	37.9	+2.85	Low
429	Flight Attendants	37.7	+2.71	Low
430	Zoologists and Wildlife Biologists	37.4	+7.37	Low
431	Furnace, Kiln, Oven, Drier, and Kettle Operators and Tenders	37.0	+0.03	Low
432	Helpers–Extraction Workers	36.8	-0.22	Low
433	Funeral Attendants	36.8	-0.23	Low
434	Surgical Technologists	35.9	+1.85	Low
435	Medical Assistants	35.8	+5.78	Low
436	Film and Video Editors	35.8	+4.76	Low
437	Cleaners of Vehicles and Equipment	35.8	-1.24	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
438	Plumbers, Pipefitters, and Steamfitters	35.7	+0.68	Low
439	Occupational Health and Safety Technicians	33.3	+8.21	Low
440	Wind Energy Operations Managers	32.7	+7.67	Low
441	Education Administrators, Kindergarten through Secondary	32.5	+7.00	Low
442	Pediatricians, General	32.4	+8.17	Low
443	Wholesale and Retail Buyers, Except Farm Products	32.2	+3.18	Low
444	Compliance Managers	32.1	+7.06	Low
445	Career/Technical Education Teachers, Middle School	32.0	+6.00	Low
446	Loss Prevention Managers	31.8	+6.84	Low
447	Probation Officers and Correctional Treatment Specialists	31.7	+6.74	Low
448	Security Managers	31.7	+6.48	Low
449	Information Security Engineers	31.6	+8.29	Low
450	Regulatory Affairs Managers	31.5	+6.50	Low
451	Clinical Data Managers	31.4	+8.29	Low
452	Health Informatics Specialists	31.3	+7.84	Low
453	Hydrologic Technicians	31.2	+7.43	Low
454	Brownfield Redevelopment Specialists and Site Managers	31.2	+6.23	Low
455	Business Continuity Planners	31.2	+8.23	Low
456	Document Management Specialists	31.2	+7.84	Low
457	Occupational Health and Safety Specialists	31.2	+7.42	Low
458	Calibration Technologists and Technicians	31.2	+7.58	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
459	Disc Jockeys, Except Radio	31.1	+6.41	Low
460	First-Line Supervisors of Retail Sales Workers	31.1	+3.10	Low
461	Lighting Technicians	31.1	+6.30	Low
462	Medical Dosimetrists	31.0	+6.68	Low
463	Web and Digital Interface Designers	31.0	+6.55	Low
464	Geographers	31.0	+6.01	Low
465	Nurse Practitioners	31.0	+7.25	Low
466	Non-Destructive Testing Specialists	30.9	+6.95	Low
467	Facilities Managers	30.9	+7.01	Low
468	Special Education Teachers, Kindergarten	30.9	+6.10	Low
469	First-Line Supervisors of Material-Moving Machine and Vehicle Operators	30.9	+7.65	Low
470	Bioinformatics Technicians	30.7	+7.68	Low
471	Precision Agriculture Technicians	30.7	+6.44	Low
472	Captains, Mates, and Pilots of Water Vessels	30.7	+3.73	Low
473	Environmental Science and Protection Technicians, Including Health	30.7	+7.27	Low
474	Special Education Teachers, Secondary School	30.7	+7.13	Low
475	Computer Systems Engineers/Architects	30.7	+7.26	Low
476	Database Administrators	30.7	+6.97	Low
477	Paramedics	30.6	+6.57	Low
478	Psychiatrists	30.6	+5.80	Low
479	Blockchain Engineers	30.6	+7.28	Low
480	Computer User Support Specialists	30.6	+7.41	Low
481	School Bus Monitors	30.6	+5.53	Low
482	Nurse Anesthetists	30.6	+5.34	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
483	Environmental Engineering Technologists and Technicians	30.5	+5.50	Low
484	Advanced Practice Psychiatric Nurses	30.5	+5.26	Low
485	Web Administrators	30.5	+7.55	Low
486	Information Security Analysts	30.5	+8.56	Low
487	Radiologists	30.5	+7.12	Low
488	Digital Forensics Analysts	30.5	+7.57	Low
489	Entertainment and Recreation Managers, Except Gambling	30.5	+7.20	Low
490	Clinical and Counseling Psychologists	30.5	+5.71	Low
491	Medical Records Specialists	30.4	+7.29	Low
492	Health Information Technologists and Medical Registrars	30.4	+7.43	Low
493	Cardiovascular Technologists and Technicians	30.4	+7.40	Low
494	Family Medicine Physicians	30.4	+7.87	Low
495	Computer Science Teachers, Postsecondary	30.4	+8.42	Low
496	Pediatric Surgeons	30.3	+6.02	Low
497	Computer Network Support Specialists	30.3	+6.20	Low
498	General Internal Medicine Physicians	30.3	+6.82	Low
499	Art Therapists	30.2	+5.36	Low
500	Appraisers of Personal and Business Property	30.2	+7.38	Low
501	Computer Systems Analysts	30.1	+7.85	Low
502	Physical Medicine and Rehabilitation Physicians	30.1	+6.26	Low
503	Survey Researchers	30.1	+7.12	Low
504	School Psychologists	30.1	+6.96	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
505	Wind Energy Development Managers	30.1	+5.06	Low
506	Taxi Drivers	30.1	+6.40	Low
507	Security Management Specialists	30.0	+7.04	Low
508	Spa Managers	30.0	+6.18	Low
509	Computer Network Architects	30.0	+6.95	Low
510	Aircraft Service Attendants	30.0	+5.36	Low
511	Appraisers and Assessors of Real Estate	29.9	+5.76	Low
512	Orthopedic Surgeons, Except Pediatric	29.8	+6.02	Low
513	Emergency Medical Technicians	29.7	+6.27	Low
514	Special Education Teachers, Elementary School	29.7	+6.11	Low
515	Financial Risk Specialists	29.7	+6.30	Low
516	Medical Equipment Repairers	29.7	+2.69	Low
517	Explosives Workers, Ordnance Handling Experts, and Blasters	29.7	+4.16	Low
518	Funeral Home Managers	29.6	+5.79	Low
519	First-Line Supervisors of Security Workers	29.6	+4.91	Low
520	Clinical Nurse Specialists	29.6	+6.21	Low
521	Penetration Testers	29.6	+6.01	Low
522	Health Specialties Teachers, Postsecondary	29.6	+7.10	Low
523	Computer Hardware Engineers	29.5	+7.54	Low
524	Acute Care Nurses	29.5	+5.91	Low
525	Allergists and Immunologists	29.5	+5.64	Low
526	Special Education Teachers, Preschool	29.5	+4.54	Low
527	Forestry and Conservation Science Teachers, Postsecondary	29.5	+7.38	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
528	Hospitalists	29.5	+6.60	Low
529	Exercise Physiologists	29.5	+6.09	Low
530	Sports Medicine Physicians	29.4	+5.53	Low
531	Obstetricians and Gynecologists	29.4	+6.30	Low
532	Biostatisticians	29.4	+7.41	Low
533	Career/Technical Education Teachers, Postsecondary	29.4	+5.55	Low
534	Nursing Instructors and Teachers, Postsecondary	29.4	+6.52	Low
535	Fundraising Managers	29.4	+5.92	Low
536	Neurologists	29.4	+5.86	Low
537	Special Education Teachers, Middle School	29.4	+6.10	Low
538	Geological Technicians, Except Hydrologic Technicians	29.4	+5.61	Low
539	Neurodiagnostic Technologists	29.3	+5.85	Low
540	Architecture Teachers, Postsecondary	29.3	+7.26	Low
541	Librarians and Media Collections Specialists	29.3	+5.65	Low
542	Statisticians	29.3	+7.25	Low
543	Registered Nurses	29.2	+5.67	Low
544	Fitness and Wellness Coordinators	29.2	+6.44	Low
545	Music Therapists	29.2	+3.84	Low
546	Business Teachers, Postsecondary	29.2	+6.98	Low
547	Software Quality Assurance Analysts and Testers	29.2	+7.68	Low
548	Recycling Coordinators	29.1	+4.62	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
549	First-Line Supervisors of Entertainment and Recreation Workers, Except Gambling Services	29.1	+6.11	Low
550	First-Line Supervisors of Passenger Attendants	29.1	+6.27	Low
551	Financial and Investment Analysts	29.1	+6.11	Low
552	Solar Sales Representatives and Assessors	29.1	+4.11	Low
553	Psychology Teachers, Postsecondary	29.1	+6.97	Low
554	Database Architects	29.1	+5.20	Low
555	Communications Teachers, Postsecondary	29.1	+6.53	Low
556	Ophthalmologists, Except Pediatric	29.1	+5.07	Low
557	Agricultural Sciences Teachers, Postsecondary	29.1	+6.97	Low
558	Sales Representatives, Wholesale and Manufacturing, Technical and Scientific Products	29.0	+4.04	Low
559	Physics Teachers, Postsecondary	29.0	+6.96	Low
560	Nurse Midwives	29.0	+5.38	Low
561	Tutors	29.0	+6.67	Low
562	Biological Science Teachers, Postsecondary	28.9	+6.52	Low
563	Midwives	28.9	+5.35	Low
564	Engineering Teachers, Postsecondary	28.9	+6.97	Low
565	Crematory Operators	28.9	+4.99	Low
566	Data Scientists	28.9	+6.59	Low
567	Psychiatric Aides	28.9	+4.34	Low
568	Critical Care Nurses	28.9	+5.60	Low
569	Legislators	28.9	+5.71	Low
570	Speech-Language Pathology Assistants	28.8	+5.80	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
571	Skincare Specialists	28.8	-0.17	Low
572	Fundraisers	28.8	+5.55	Low
573	Library Science Teachers, Postsecondary	28.8	+6.67	Low
574	Geothermal Technicians	28.8	+4.32	Low
575	Emergency Medicine Physicians	28.8	+5.21	Low
576	Mathematical Science Teachers, Postsecondary	28.8	+6.67	Low
577	Recreation and Fitness Studies Teachers, Postsecondary	28.8	+6.53	Low
578	Family and Consumer Sciences Teachers, Postsecondary	28.7	+6.97	Low
579	Sales Representatives of Services, Except Advertising, Insurance, Financial Services, and Travel	28.6	+5.56	Low
580	Dermatologists	28.6	+4.19	Low
581	Software Developers	28.6	+6.52	Low
582	Chemistry Teachers, Postsecondary	28.6	+6.52	Low
583	Environmental Science Teachers, Postsecondary	28.6	+6.83	Low
584	Orthoptists	28.6	+4.50	Low
585	Physicians, Pathologists	28.6	+5.77	Low
586	Naturopathic Physicians	28.5	+5.41	Low
587	Data Warehousing Specialists	28.5	+6.95	Low
588	Business Intelligence Analysts	28.5	+6.71	Low
589	Human Resources Specialists	28.5	+6.39	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
590	Network and Computer Systems Administrators	28.5	+6.66	Low
591	Computer Programmers	28.5	+6.08	Low
592	News Analysts, Reporters, and Journalists	28.4	+6.52	Low
593	Computer and Information Research Scientists	28.4	+6.27	Low
594	Law Teachers, Postsecondary	28.4	+6.26	Low
595	Education Teachers, Postsecondary	28.4	+6.26	Low
596	History Teachers, Postsecondary	28.4	+6.21	Low
597	First-Line Supervisors of Gambling Services Workers	28.3	+4.46	Low
598	Social Work Teachers, Postsecondary	28.3	+6.10	Low
599	Acupuncturists	28.3	+5.05	Low
600	Retail Loss Prevention Specialists	28.2	+3.43	Low
601	Athletes and Sports Competitors	28.2	+0.23	Low
602	Atmospheric, Earth, Marine, and Space Sciences Teachers, Postsecondary	28.2	+6.24	Low
603	English Language and Literature Teachers, Postsecondary	28.2	+5.78	Low
604	Sociology Teachers, Postsecondary	28.2	+6.52	Low
605	Gambling Managers	28.2	+4.17	Low
606	Radiologic Technologists and Technicians	28.2	+4.75	Low
607	Patient Representatives	28.1	+5.28	Low
608	Art, Drama, and Music Teachers, Postsecondary	28.1	+5.22	Low
609	Economics Teachers, Postsecondary	28.1	+6.22	Low
610	Urologists	28.1	+4.98	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
611	Criminal Justice and Law Enforcement Teachers, Postsecondary	28.1	+5.67	Low
612	Wind Turbine Service Technicians	28.1	+2.75	Low
613	Cooks, Private Household	28.1	-1.94	Low
614	Food Science Technicians	28.1	+6.05	Low
615	Sustainability Specialists	28.1	+5.05	Low
616	Cardiologists	28.0	+5.66	Low
617	Preventive Medicine Physicians	28.0	+5.26	Low
618	Area, Ethnic, and Cultural Studies Teachers, Postsecondary	27.9	+5.67	Low
619	Information Technology Project Managers	27.9	+5.50	Low
620	Anthropology and Archeology Teachers, Postsecondary	27.9	+5.65	Low
621	Surgical Assistants	27.9	+3.19	Low
622	Geography Teachers, Postsecondary	27.9	+6.25	Low
623	Genetic Counselors	27.8	+5.60	Low
624	Labor Relations Specialists	27.8	+4.22	Low
625	Anesthesiologists	27.8	+4.69	Low
626	Agents and Business Managers of Artists, Performers, and Athletes	27.7	+3.74	Low
627	Adapted Physical Education Specialists	27.7	+3.77	Low
628	Photonics Technicians	27.7	+3.71	Low
629	Project Management Specialists	27.7	+4.98	Low
630	Geographic Information Systems Technologists and Technicians	27.7	+5.63	Low
631	Web Developers	27.6	+4.75	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
632	Community Health Workers	27.6	+5.34	Low
633	Aircraft Cargo Handling Supervisors	27.6	+4.22	Low
634	Occupational Therapy Aides	27.6	+0.55	Low
635	Foreign Language and Literature Teachers, Postsecondary	27.5	+5.36	Low
636	Video Game Designers	27.4	+5.07	Low
637	First-Line Supervisors of Helpers, Laborers, and Material Movers, Hand	27.4	+5.03	Low
638	Magnetic Resonance Imaging Technologists	27.4	+4.39	Low
639	Teaching Assistants, Preschool, Elementary, Middle, and Secondary School, Except Special Education	27.3	+4.79	Low
640	Philosophy and Religion Teachers, Postsecondary	27.3	+5.08	Low
641	Political Science Teachers, Postsecondary	27.0	+4.49	Low
642	Hearing Aid Specialists	26.6	+2.40	Low
643	Online Merchants	26.6	+3.57	Low
644	Forest and Conservation Technicians	26.6	+2.61	Low
645	Computer Numerically Controlled Tool Operators	26.4	+2.68	Low
646	Teaching Assistants, Special Education	26.1	+1.88	Low
647	Actuaries	26.1	+5.05	Low
648	Computer Numerically Controlled Tool Programmers	26.0	+3.71	Low
649	First-Line Supervisors of Personal Service Workers	26.0	+3.18	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
650	Court Reporters and Simultaneous Captioners	25.8	+4.87	Low
651	Solar Photovoltaic Installers	25.8	+1.94	Low
652	Weatherization Installers and Technicians	25.7	+3.12	Low
653	Morticians, Undertakers, and Funeral Arrangers	25.6	+0.67	Low
654	Epidemiologists	25.6	+5.55	Low
655	Farmworkers and Laborers, Crop, Nursery, and Greenhouse	25.5	+0.77	Low
656	Telecommunications Engineering Specialists	25.4	+2.03	Low
657	Substitute Teachers, Short-Term	25.3	+3.46	Low
658	Transportation Security Screeners	25.2	+0.64	Low
659	Fishing and Hunting Workers	24.9	+1.51	Low
660	Agricultural Technicians	24.7	+1.51	Low
661	Ophthalmic Medical Technicians	24.7	+1.86	Low
662	Airline Pilots, Copilots, and Flight Engineers	24.7	+6.67	Low
663	Ophthalmic Medical Technologists	24.6	+2.39	Low
664	Shuttle Drivers and Chauffeurs	24.2	+1.51	Low
665	Ambulance Drivers and Attendants, Except Emergency Medical Technicians	24.2	-0.82	Low
666	Stockers and Order Fillers	24.1	-0.11	Low
667	Bus Drivers, School	24.1	+2.68	Low
668	Teaching Assistants, Postsecondary	24.1	+2.84	Low
669	Adult Basic Education, Adult Secondary Education, and English as a Second Language Instructors	24.0	+5.04	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
670	Farmworkers, Farm, Ranch, and Aquacultural Animals	23.9	+0.78	Low
671	Petroleum Engineers	23.9	+7.85	Low
672	Endoscopy Technicians	23.8	+2.25	Low
673	Railroad Brake, Signal, and Switch Operators and Locomotive Firers	23.8	+1.36	Low
674	Solar Energy Installation Managers	23.8	+6.77	Low
675	Phlebotomists	23.7	+1.94	Low
676	Middle School Teachers, Except Special and Career/Technical Education	23.5	+6.52	Low
677	Financial Examiners	23.5	+6.52	Low
678	Roof Bolters, Mining	23.4	-0.50	Low
679	Coating, Painting, and Spraying Machine Setters, Operators, and Tenders	23.4	+0.02	Low
680	Excavating and Loading Machine and Dragline Operators, Surface Mining	23.2	-0.64	Low
681	Loading and Moving Machine Operators, Underground Mining	23.1	+0.62	Low
682	Agricultural Equipment Operators	23.0	-0.07	Low
683	Nursing Assistants	23.0	-0.04	Low
684	Home Health Aides	22.9	+0.51	Low
685	Earth Drillers, Except Oil and Gas	22.9	-0.41	Low
686	Orderlies	22.8	-1.08	Low
687	Baristas	22.8	+0.50	Low
688	Timing Device Assemblers and Adjusters	22.8	+0.36	Low
689	Bus Drivers, Transit and Intercity	22.3	-0.09	Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
690	Public Relations Specialists	22.3	+4.30	Low
691	Commercial Divers	21.9	+3.85	Low
692	Fast Food and Counter Workers	21.7	-1.23	Low
693	Animal Control Workers	21.5	+0.54	Low
694	Personal Care Aides	21.5	-0.48	Low
695	Anesthesiologist Assistants	21.4	+7.41	Low
696	Fiberglass Laminators and Fabricators	21.3	-0.41	Low
697	Dental Hygienists	20.9	-0.91	Low
698	Concierges	20.9	-0.08	Low
699	Desktop Publishers	20.7	+4.74	Low
700	Physician Assistants	20.6	+6.65	Low
701	Kindergarten Teachers, Except Special Education	20.6	+5.58	Low
702	Mining and Geological Engineers, Including Mining Safety Engineers	20.6	+6.57	Low
703	Optometrists	20.1	+6.08	Low
704	First-Line Supervisors of Construction Trades and Extraction Workers	20.0	+3.01	Low
705	Nuclear Medicine Technologists	20.0	+6.97	Very Low
706	Urban and Regional Planners	19.8	+6.80	Very Low
707	Management Analysts	19.5	+6.53	Very Low
708	Firefighters	19.5	+2.47	Very Low
709	Manufactured Building and Mobile Home Installers	18.9	+0.87	Very Low
710	Social and Human Service Assistants	18.4	+5.35	Very Low
711	Air Traffic Controllers	18.1	+7.10	Very Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
712	Electrical Engineers	17.7	+7.71	Very Low
713	Electricians	17.6	+2.58	Very Low
714	Chemists	17.4	+7.41	Very Low
715	Customs and Border Protection Officers	17.0	+7.22	Very Low
716	Physicists	16.2	+6.23	Very Low
717	Police and Sheriff's Patrol Officers	16.0	+6.18	Very Low
718	Environmental Compliance Inspectors	15.7	+7.69	Very Low
719	Regulatory Affairs Specialists	15.5	+7.48	Very Low
720	Food Scientists and Technologists	15.3	+7.63	Very Low
721	Equal Opportunity Representatives and Officers	15.1	+7.08	Very Low
722	Financial Managers	15.0	+8.07	Very Low
723	Government Property Inspectors and Investigators	14.8	+6.80	Very Low
724	Treasurers and Controllers	14.7	+7.79	Very Low
725	Customs Brokers	14.6	+6.64	Very Low
726	Sound Engineering Technicians	14.6	+1.58	Very Low
727	Compliance Officers	14.5	+6.46	Very Low
728	Travel Agents	14.4	+4.49	Very Low
729	Broadcast Announcers and Radio Disc Jockeys	14.3	+4.35	Very Low
730	Dietetic Technicians	13.7	+0.71	Very Low
731	Nuclear Engineers	13.5	+6.50	Very Low
732	Fish and Game Wardens	13.3	+5.35	Very Low
733	First-Line Supervisors of Non-Retail Sales Workers	13.3	+5.81	Very Low
734	General and Operations Managers	13.1	+4.04	Very Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
735	Investment Fund Managers	13.0	+6.11	Very Low
736	Chefs and Head Cooks	13.0	+2.97	Very Low
737	Self-Enrichment Teachers	12.9	-0.08	Very Low
738	Construction Managers	12.9	+5.80	Very Low
739	Respiratory Therapists	12.8	+6.24	Very Low
740	Food Service Managers	12.7	+4.42	Very Low
741	Graphic Designers	12.7	+4.47	Very Low
742	Coroners	12.4	+4.40	Very Low
743	Animal Scientists	12.4	+6.28	Very Low
744	Computer and Information Systems Managers	12.3	+8.85	Very Low
745	Hairdressers, Hairstylists, and Cosmetologists	12.3	+1.25	Very Low
746	Dancers	11.6	-1.43	Very Low
747	Electrical Power-Line Installers and Repairers	11.5	+1.84	Very Low
748	Arbitrators, Mediators, and Conciliators	11.5	+5.50	Very Low
749	Sociologists	11.5	+5.56	Very Low
750	Licensed Practical and Licensed Vocational Nurses	11.3	+5.52	Very Low
751	Bioengineers and Biomedical Engineers	11.3	+7.57	Very Low
752	Farmers, Ranchers, and Other Agricultural Managers	11.2	+6.49	Very Low
753	Health Education Specialists	11.1	+6.56	Very Low
754	Validation Engineers	10.9	+7.98	Very Low
755	Biomass Power Plant Managers	10.8	+7.84	Very Low
756	Transportation Planners	10.8	+6.79	Very Low
757	Astronomers	10.8	+6.68	Very Low
758	Purchasing Managers	10.7	+7.71	Very Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
759	Aerospace Engineers	10.7	+9.00	Very Low
760	Industrial Engineering Technologists and Technicians	10.7	+7.68	Very Low
761	Environmental Scientists and Specialists, Including Health	10.7	+7.37	Very Low
762	Editors	10.6	+5.08	Very Low
763	Transportation Engineers	10.6	+8.67	Very Low
764	Environmental Restoration Planners	10.6	+7.26	Very Low
765	Health and Safety Engineers, Except Mining Safety Engineers and Inspectors	10.5	+7.68	Very Low
766	Manufacturing Engineers	10.5	+7.56	Very Low
767	Nanotechnology Engineering Technologists and Technicians	10.4	+7.41	Very Low
768	Biofuels Production Managers	10.4	+7.40	Very Low
769	Hydroelectric Production Managers	10.4	+7.36	Very Low
770	Radio Frequency Identification Device Specialists	10.3	+7.84	Very Low
771	Mathematicians	10.3	+5.62	Very Low
772	Nannies	10.3	+1.90	Very Low
773	Lawyers	10.2	+6.67	Very Low
774	Bioinformatics Scientists	9.9	+8.43	Very Low
775	Molecular and Cellular Biologists	9.9	+8.41	Very Low
776	Fire-Prevention and Protection Engineers	9.9	+7.11	Very Low
777	Political Scientists	9.8	+5.93	Very Low
778	Advertising and Promotions Managers	9.8	+5.93	Very Low
779	Meeting, Convention, and Event Planners	9.7	+6.04	Very Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
780	Solar Energy Systems Engineers	9.7	+8.29	Very Low
781	Landscape Architects	9.7	+5.15	Very Low
782	Veterinarians	9.7	+5.85	Very Low
783	Environmental Engineers	9.6	+7.83	Very Low
784	Prosthodontists	9.6	+4.09	Very Low
785	Human Factors Engineers and Ergonomists	9.6	+6.68	Very Low
786	Geothermal Production Managers	9.5	+6.53	Very Low
787	Quality Control Systems Managers	9.5	+6.52	Very Low
788	Civil Engineers	9.5	+7.58	Very Low
789	Geneticists	9.5	+7.97	Very Low
790	Media Programming Directors	9.5	+7.27	Very Low
791	Water/Wastewater Engineers	9.4	+7.48	Very Low
792	Marine Engineers and Naval Architects	9.3	+8.29	Very Low
793	Conservation Scientists	9.2	+7.55	Very Low
794	Industrial Ecologists	9.1	+5.80	Very Low
795	Water Resource Specialists	9.1	+7.25	Very Low
796	Art Directors	9.0	+6.69	Very Low
797	Industrial Engineers	9.0	+6.07	Very Low
798	Producers and Directors	9.0	+6.77	Very Low
799	Residential Advisors	9.0	+2.56	Very Low
800	Architectural and Engineering Managers	9.0	+7.26	Very Low
801	Animal Trainers	8.9	-1.09	Very Low
802	Industrial Production Managers	8.9	+5.88	Very Low
803	Climate Change Policy Analysts	8.8	+5.53	Very Low
804	Pharmacists	8.8	+7.55	Very Low
805	Credit Counselors	8.7	+4.69	Very Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
806	Robotics Engineers	8.7	+7.28	Very Low
807	First-Line Supervisors of Office and Administrative Support Workers	8.7	+7.27	Very Low
808	Energy Engineers, Except Wind and Solar	8.7	+7.27	Very Low
809	Microsystems Engineers	8.7	+7.25	Very Low
810	Curators	8.7	+7.97	Very Low
811	Nanosystems Engineers	8.6	+7.24	Very Low
812	Soil and Plant Scientists	8.6	+6.52	Very Low
813	Range Managers	8.6	+6.96	Very Low
814	First-Line Supervisors of Production and Operating Workers	8.6	+6.95	Very Low
815	Physical Therapists	8.5	+6.42	Very Low
816	Forensic Science Technicians	8.5	+7.55	Very Low
817	Training and Development Managers	8.5	+7.85	Very Low
818	Travel Guides	8.5	+2.77	Very Low
819	Operations Research Analysts	8.4	+4.95	Very Low
820	Hydrologists	8.4	+7.02	Very Low
821	Forest Fire Inspectors and Prevention Specialists	8.4	+3.58	Very Low
822	Biologists	8.3	+6.82	Very Low
823	Logistics Analysts	8.3	+7.11	Very Low
824	Materials Engineers	8.3	+6.21	Very Low
825	First-Line Supervisors of Correctional Officers	8.3	+5.79	Very Low
826	Orthodontists	8.3	+5.98	Very Low
827	Microbiologists	8.2	+6.96	Very Low
828	Logistics Engineers	8.2	+6.96	Very Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
829	Biochemists and Biophysicists	8.1	+5.40	Very Low
830	Automotive Engineers	8.1	+6.99	Very Low
831	Electronics Engineers, Except Computer	8.1	+5.56	Very Low
832	Medical and Health Services Managers	8.0	+7.28	Very Low
833	Photonics Engineers	8.0	+6.56	Very Low
834	Securities, Commodities, and Financial Services Sales Agents	8.0	+6.35	Very Low
835	Psychiatric Technicians	7.9	+3.64	Very Low
836	Secondary School Teachers, Except Special and Career/Technical Education	7.9	+7.11	Very Low
837	Natural Sciences Managers	7.9	+6.08	Very Low
838	Commercial and Industrial Designers	7.9	+4.18	Very Low
839	Farm and Home Management Educators	7.9	+7.12	Very Low
840	Ship Engineers	7.9	+3.77	Very Low
841	Exercise Trainers and Group Fitness Instructors	7.8	-0.65	Very Low
842	Interior Designers	7.8	+5.60	Very Low
843	Mechatronics Engineers	7.8	+6.36	Very Low
844	Chief Executives	7.8	+6.25	Very Low
845	Clinical Research Coordinators	7.7	+5.94	Very Low
846	Child, Family, and School Social Workers	7.7	+4.93	Very Low
847	Chiropractors	7.7	+4.98	Very Low
848	Poets, Lyricists and Creative Writers	7.7	+3.87	Very Low
849	Mechanical Engineers	7.7	+6.57	Very Low
850	Lodging Managers	7.7	+7.26	Very Low
851	First-Line Supervisors of Police and Detectives	7.7	+7.21	Very Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
852	Musicians and Singers	7.6	+0.23	Very Low
853	Substance Abuse and Behavioral Disorder Counselors	7.6	+4.32	Very Low
854	Media Technical Directors/Managers	7.6	+5.41	Very Low
855	Public Relations Managers	7.6	+6.10	Very Low
856	Elementary School Teachers, Except Special Education	7.5	+7.11	Very Low
857	Education Administrators, Postsecondary	7.5	+6.53	Very Low
858	Marketing Managers	7.5	+6.09	Very Low
859	Social and Community Service Managers	7.5	+6.82	Very Low
860	Materials Scientists	7.5	+5.36	Very Low
861	Dietitians and Nutritionists	7.4	+7.04	Very Low
862	Instructional Coordinators	7.3	+6.91	Very Low
863	Fuel Cell Engineers	7.3	+6.22	Very Low
864	Rehabilitation Counselors	7.3	+6.37	Very Low
865	Architects, Except Landscape and Naval	7.3	+5.49	Very Low
866	Medical Scientists, Except Epidemiologists	7.3	+6.81	Very Low
867	Biofuels/Biodiesel Technology and Product Development Managers	7.2	+5.54	Very Low
868	Emergency Management Directors	7.2	+6.92	Very Low
869	Childcare Workers	7.2	-1.19	Very Low
870	Orthotists and Prosthetists	7.2	+6.85	Very Low
871	Directors, Religious Activities and Education	7.2	+4.65	Very Low
872	Wind Energy Engineers	7.1	+5.66	Very Low
873	Education and Childcare Administrators, Preschool and Daycare	7.0	+6.54	Very Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
874	Career/Technical Education Teachers, Secondary School	7.0	+6.10	Very Low
875	Neuropsychologists	6.9	+6.51	Very Low
876	First-Line Supervisors of Firefighting and Prevention Workers	6.8	+6.47	Very Low
877	Marriage and Family Therapists	6.8	+5.41	Very Low
878	Speech-Language Pathologists	6.8	+6.14	Very Low
879	Chief Sustainability Officers	6.7	+5.21	Very Low
880	Sales Managers	6.7	+5.41	Very Low
881	Writers and Authors	6.7	+2.87	Very Low
882	Human Resources Managers	6.7	+6.10	Very Low
883	Industrial-Organizational Psychologists	6.7	+5.45	Very Low
884	Audiologists	6.6	+6.30	Very Low
885	Park Naturalists	6.5	+4.92	Very Low
886	Anthropologists and Archeologists	6.4	+5.66	Very Low
887	Veterinary Technologists and Technicians	6.4	+3.51	Very Low
888	First-Line Supervisors of Mechanics, Installers, and Repairers	6.4	+6.08	Very Low
889	Special Effects Artists and Animators	6.3	+4.81	Very Low
890	Sales Engineers	6.3	+5.84	Very Low
891	Educational, Guidance, and Career Counselors and Advisors	6.2	+5.30	Very Low
892	Fashion Designers	6.1	+4.03	Very Low
893	Talent Directors	6.1	+3.92	Very Low
894	Logisticians	6.1	+4.92	Very Low
895	Dentists, General	6.1	+5.62	Very Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
896	Coaches and Scouts	5.9	+4.63	Very Low
897	Chemical Engineers	5.9	+4.22	Very Low
898	Occupational Therapists	5.8	+5.43	Very Low
899	Clinical Neuropsychologists	5.8	+5.34	Very Low
900	Training and Development Specialists	5.7	+4.34	Very Low
901	Mental Health Counselors	5.7	+5.19	Very Low
902	Podiatrists	5.6	+5.13	Very Low
903	Foresters	5.5	+4.71	Very Low
904	Floral Designers	5.5	+0.79	Very Low
905	Clergy	5.3	+4.48	Very Low
906	Recreational Therapists	5.2	+4.91	Very Low
907	Physical Therapist Assistants	5.1	+3.31	Very Low
908	Low Vision Therapists, Orientation and Mobility Specialists, and Vision Rehabilitation Therapists	5.0	+4.62	Very Low
909	Oral and Maxillofacial Surgeons	4.8	+4.49	Very Low
910	Athletic Trainers	4.8	+4.06	Very Low
911	Occupational Therapy Assistants	4.7	+1.89	Very Low
912	Music Directors and Composers	4.6	+3.14	Very Low
913	Set and Exhibit Designers	4.6	+4.05	Very Low
914	Mental Health and Substance Abuse Social Workers	4.3	+4.04	Very Low
915	Fine Artists, Including Painters, Sculptors, and Illustrators	4.3	+0.07	Very Low
916	Healthcare Social Workers	4.2	+3.89	Very Low
917	Craft Artists	3.1	-0.35	Very Low

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Table B.1 continued from previous page

#	Occupation	MAEI 2026	Delta	Risk Tier
918	Recreation Workers	2.7	+2.13	Very Low
919	Photographers	2.7	+0.56	Very Low
920	Choreographers	2.0	+1.56	Very Low
921	Preschool Teachers, Except Special Education	1.2	+0.43	Very Low
922	Fabric and Apparel Patternmakers	0.4	-0.06	Very Low
923	Makeup Artists, Theatrical and Performance	0.4	-0.61	Very Low

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